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Digital Divide

CIVIC ENGAGEMENT, INFORMATION POVERTY, AND THE INTERNET WORLDWIDE

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CHAPTER 1

The Digital Divide

The year 1989 dawned like any other but, in retrospect, it witnessed two major developments of immense historical significance. One was highly visible and widely celebrated: the symbolic dismantling of the Berlin Wall sparking the brushfire of electoral democracy spreading throughout the post-Communist world and beyond. The other was less generally recognized at the time, beyond a few scientific and technical cognoscenti: the invention of the World Wide Web. Dispersed computers communicating via packet-switching networks, and hence a rudimentary version of the Internet, had linked scientific elites for two decades. It took the invention of the Web by Tim Berners-Lee in CERN and the launch of a graphical browser, Mosaic, four years later to popularize this technology. Like a stone dropping into a pellucid pond, the ripples from this invention are surging throughout industrialized societies at the core, as well as flowing more slowly among developing societies at the periphery. With the size of the online community doubling every year, few doubt the potential importance of the Internet for transforming the way people live, work, and play. But, beyond these spheres, what are the causes of stratification in the networked world? In particular – the core focus of this book – will the Internet serve to reinforce or erode the gap between information-rich and poor nations? Will it exacerbate or reduce social divisions within countries? And will it strengthen representative democracy, as many hope, or will it buttress the power of established interests, as others fear?

In exploring these issues, this book focuses on understanding the root causes and the major consequences of inequalities evident during the first decade of the Internet age. The term "digital divide" has quickly become so popular as an instant sound bite that it has entered everyday

speech as shorthand for any and every disparity within the online community. In this study the concept of the digital divide is understood as a multidimensional phenomenon encompassing three distinct aspects. The *global divide* refers to the divergence of Internet access between industrialized and developing societies. The *social divide* concerns the gap between information rich and poor in each nation. And finally within the online community, the *democratic divide* signifies the difference between those who do, and do not, use the panoply of digital resources to engage, mobilize, and participate in public life. To consider these matters, this introduction summarizes the contemporary debate about these issues, and then outlines the book's central argument, framework, and organization.

THE GLOBAL DIVIDE AMONG COUNTRIES

Few doubt the potential impact of digital technologies for reshaping the flow of investment, goods, and services in the global marketplace. Like the Californian Gold Rush of the 1850s, dot.coms have scrambled to stake their claims in the virtual frontier. Productivity and efficiency gains from investments in ICTs remain difficult to gauge but the U.S. Department of Commerce estimates that industries producing computer and communications hardware, software, and services have had a major impact on the U.S. economy.[1] These developments fueled an intense flurry of heady speculation about the emergence of a "new" economy breaking the traditional business rules, although, mirroring the fluctuating fortunes of the Nasdaq index and the death of hundreds of dot.com start-ups, more cautious voices have subsequently warned that beyond a few isolated sectors, such as the travel or insurance industries, "bricks and mortar" assets still count for successful business-customer relations, along with old-fashioned notions such as profitability for investors, brand names, sales, and distribution systems.[2]

In the social sphere, few question the significance of cyberculture for transforming leisure hours, community networks, and personal lifestyles.[3] Thousands of Internet sites and over 2 billion web pages cater to every conceivable interest from acupuncture to zoology.[4] Within a decade of its launch, America has become all Internet, all the time. The public has also flooded online in comparable countries such as Canada, Sweden, and Australia.[5] The Internet population surged from about 3

million worldwide users in 1994 to more than 400 million in late-2000.[6] Yet the potential for this medium, currently reaching about 7 percent of the world's population, has only started to be exploited. Despite some indications of a possible slowdown in sales of personal computers in the saturated U.S. market, connectivity seems likely to gain momentum in the near future: Metcalf's law suggests that the value of a network is proportional to the square number of people using it: the more people link to the Internet, the greater its utility, the more it attracts.[7]

But what has been, and what will be, the impact of digital technologies on poorer countries? Surf at random, click on this, click on that, and whose voices do you hear around the globe? There are many plausible reasons why the emerging Internet age may reinforce disparities between postindustrial economies at the core of the network and developing societies at the periphery.[8] As many warn, the basic problem is "To them that hath shall be given". If investment in digital technologies has the capacity to boost productivity, advanced economies such as Sweden, Australia, and the United States at the forefront of the technological revolution may be well placed to pull even farther ahead, maintaining their edge in future decades. A few middle-level economies like Taiwan, Brazil, and South Korea may manage to leverage themselves profitably into niche markets within the global marketplace, servicing international corporations based elsewhere by providing software development or manufacturing silicon chips. But most poorer societies, lagging far behind, plagued by multiple burdens of debt, disease, and ignorance, may join the digital world decades later and, in the long-term, may ultimately fail to catch up.[9]

International organizations have sounded the alarm. The OECD warns that affluent states at the cutting edge of technological change have reinforced their lead in the new knowledge economy but so far the benefits of the Internet have not yet trickled down far to Southern, Central, and Eastern Europe, let alone to the poorest areas in Sub-Saharan Africa, Latin America, and Southeast Asia.[10] The UN Development Report argues that productivity gains from information technologies may widen the chasm between the most affluent nations and those that lack the skills, resources, and infrastructure to invest in the information society: *"The network society is creating parallel communications systems: one for those with income, education and literally connections, giving plentiful information at low cost and high speed; the other for those without connections, blocked by high barriers of time, cost*

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and uncertainty and dependent upon outdated information."[11] Echoing these concerns, UNESCO emphasizes that most of the world's population lack basic access to a telephone, let alone a computer, producing societies increasingly marginalized at the periphery of communication networks.[12] Leaders in the World Bank, European Union, United Nations, and G-8 have highlighted the problems of exclusion from the knowledge economy, where know-how replaces land and capital as the basic building blocks of growth.[13] Initiatives have been launched to address this problem but disparities in the distribution of information and communication technologies are deep seated, suggesting that they will not easily be eradicated or ameliorated. The global flow of such traditional media as news, books, or scholarly research has long displayed center-periphery inequalities, with information flowing primarily from north to south; an issue generating heated debate during the 1980s centered on UNESCO's controversial New World Information Order.[14] Technology has always held promise as an engine of economic growth for transforming developing nations – including machines for printing, textiles manufacture, and iron railways in the nineteenth century, and automobiles, oil production, and television in the twentieth – but critics argue that in practice this promise has often mainly served to benefit the industrialized world.[15]

Yet at the same time *if* technological diffusion can be achieved in poorer societies, and it is a big "if," then many observers hope that the Internet provides multiple opportunities for socioeconomic and democratic development. Digital networks have the potential to broaden and enhance access to information and communications for remote rural areas and poorer neighborhoods, to strengthen the process of democratization under transitional regimes, and to ameliorate the endemic problems of poverty in the developing world. With connectivity as the umbilical cord, enthusiasts hope that the Internet will eventually serve multiple functions as the world's favorite public library, school classroom and medical database, post office and telephone, marketplace and shopping mall, channel for entertainment, culture and music, daily news resource for headlines, stocks and weather, and heterogeneous global public sphere. In the heady words of the G-8 Okinawa Charter: *"Our vision of an information society is one that better enables people to fulfill their potential and realize their aspirations. To this end we must ensure that IT serves the mutually supportive goals of creating sustainable economic growth, enhancing the*

public welfare, and fostering social cohesion, and work to fully realize its potential to strengthen democracy, increase transparency and accountability in governance, promote human rights, enhance cultural diversity, and to foster international peace and stability."[16] The Internet may allow societies to leapfrog stages of technological and industrial development. On the production side, if Bangalore companies can write software code for IBM or Microsoft, and if Costa Rica can manufacture chips for Intel, then potentially entrepreneurs can offer similar services from Malaysia, Brazil, and South Africa. The Internet encourages market globalization: small craft industries and the tourism industry in Bali or the Maldives can deal directly with customers and vacationers in New York and London, irrespective of distance, the costs of advertising, and the intermediate distribution chains of travel agents and retail businesses.[17] The Internet also offers promise for the delivery of basic social services such as education and health information across the globe, a function that may be particularly important for middle-level professionals serving their broader community.[18] Local teachers or community officials connected to the digital world in Lagos, Beijing, or Calcutta can access the same electronic journals, books, and databases as students at the Sorbonne, Oxford, or Harvard. Distance learning can widen access to training and education, via open universities in India, Africa, and Thailand, and language websites for schools.[19] Networks of hospitals and health care professionals in the Ukraine, Mozambique, and Stockholm can pool expertise and knowledge about the latest research on AIDS. Peasant farmers using village community centers can learn about storm warnings and market prices for their crops, along with employment opportunities in local towns. Where peripheral regions lack access to the traditional media, the convergence of communication technologies means that the Internet has the potential to deliver virtual local newspapers, streaming radio and television video, as well as other services.

It is hoped that within a few years many of the existing barriers to access will be overcome with the combination of technological breakthroughs, market competition, and state initiatives. Internet has usually been delivered via bulky desktop personal computers tethered to telephone wires, but multiple less expensive devices are rapidly facilitating wireless access, including NTT's DoCoMo mobile phones using I-mode in Japan, Nokia's Communicator using WAP-enabled services in Europe, and handheld personal digital assistants such as

Handspring and Palm Pilots which are popular in the United States.[20] Prototype disposable prepaid cell phones and laptops are under development, along with speech-recognition software and voice-activated Internet services. The price of hardware, software, and services has been plummeting, owing to increased competition in telecommunications combined with computer technologies' falling costs, faster speeds, and smaller microprocessors.[21] In the 1960s Intel founder Gordon Moore predicted that, for the foreseeable future, chip density, and hence computing power, would double every eighteen months while costs would remain constant. During the last thirty years "Moore's law" has proved remarkably prescient. Every eighteen months, you can get twice as much power for the same cost. Telecommunications bandwidth, the speed at which data can be moved through the phone network, is experiencing similarly dramatic improvements owing to high-speed fiber-optic cable, satellites, and wireless communication technologies, all of which can be used on the same network. There have been parallel developments with computer memory and storage devices such as rewritable CD-ROMs. In 1980, a gigabyte of storage cost several hundred thousand dollars and occupied a room. It now fits on a credit-card device that can be carried in your pocket. As well as technological innovations, public-sector initiatives in developing countries as diverse as Estonia, Costa Rica, and Bangladesh have promoted the infrastructure, skills training, and knowledge necessary to widen use of digital technologies.

The implications of these developments promise to sweep well beyond the economic sphere. Observers hope that digital technologies will shift some of the global disparities in power as well as wealth, by fostering a worldwide civic society countering the role of international agencies, strengthening the voice of the developing world, dissolving some of the boundaries of the nation-state, and reinforcing the process of democratization.[22] By directly linking political activists in different countries, and reducing the costs of communication and networking, the Internet may foster new types of mobilization by transnational advocacy networks around the world.[23] By connecting disparate social movements, coalitions can be formed that mobilize a global civic society, such as protestors concerned about the World Trade Organization meetings in Seattle and Washington, D.C., the anti-landmine campaign, the anti-sweatshop manufacture of Nike shoes, and opposition movements in Burma, linking indigenous groups in developing societies with

a diverse mélange of Norwegian environmentalists, Australian trade unionists, and European human rights organizations.[24] The Internet may facilitate the networking and mobilizing functions of NGOs working across national borders, as a countervailing force to the influence of technocratic elites and government leaders running traditional international organizations.[25] The role of the Internet may be even more important as a force for human rights, providing a global platform for opposition movements challenging autocratic regimes and military dictatorships, despite government attempts to restrict access in countries like China and Cuba.[26] Therefore many observers have emphasized that the emerging years of the Internet Age have generated substantial worldwide inequalities in access and use although, if this could be overcome, it is widely believed that digital technologies will provide multiple opportunities for development.

The role of technology has therefore fueled a debate among optimists envisaging the positive role of the Internet for transforming poverty in developing societies, skeptics who believe that new technologies alone will make little difference one way or another, and pessimists who emphasize that digital technologies will further exacerbate the existing North-South divide. This debate generates a series of questions that will be considered in this book. Today which nations around the globe are digital leaders and laggards? What explains variations across countries in Internet use, in particular is it levels of socioeconomic development, investments in human capital, the process of democratization, or something else? Does the Internet create new inequalities, or reinforce existing divisions evident for decades in the spread of old communication technologies? Attempts to move beyond speculative theorizing about these questions face major challenges. The World Wide Web remains in its adolescence; any examination of trends is limited to just a decade. Technology continues to evolve rapidly, along with its social uses, so that projected estimates are often rapidly overtaken by events. Yet, despite the need for considerable caution in weighing the available evidence, if we can establish the main drivers behind the diffusion of the Internet, and if these prove similar to the reasons behind the adoption of older forms of information technologies, then we are in a much better position to understand and predict the probable pattern of future developments, the potential consequences of the rise of the Internet age, and also the policy initiatives most likely to overcome the global divide.

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SOCIAL STRATIFICATION WITHIN COUNTRIES

Equally important, many official agencies have expressed concern about the development of a widening digital divide *within* societies. Technological opportunities are often unevenly distributed, even in nations like Australia, the United States, and Sweden at the forefront of the information society. As the Internet has become increasingly central to life, work, and play – providing job opportunities, strengthening community networks and facilitating educational advancement – it becomes even more important if certain groups and areas are systematically excluded, such as poorer neighborhoods, working-class households, or peripheral rural communities. Governments in many countries have recognized this issue and developed initiatives designed to tackle this potential problem. The EU prioritized social inclusion as one of the three key objectives when launching the e-Europe Action Plan in Lisbon in March 1999.[27] In the United States, a series of studies by the Department of Commerce, *Falling Through the Net*, have emphasized lower rates of Internet penetration among the poor.[28] The 1998 survey found that affluent households (with income of $75,000 and above) were twenty times as likely to have Internet access as those at the lowest income levels, and more than nine times as likely to have computer access.[29] In February 2000, President Clinton expressed concern about this situation and proposed a new plan to help bridge the "digital divide," offering private companies a $2 billion tax break, new teacher training programs, and the development of Community Technology Centers in low-income neighborhoods to help close the gap so that the Internet eventually becomes as ubiquitous as the availability of the telephone or television.[30] The Department of Commerce has headed this initiative, emphasizing the role of programs to widen public access, promote digital skills, and encourage content that will empower underserved communities. The most common policy strategy has been to wire classrooms, although some warn that by itself this may be insufficient to close the digital divide.[31] The survey in August 2000 found that many groups that have traditionally lacked digital opportunities have been making substantial gains in connectivity and computer ownership, with the rising Internet tide carrying many boats. Nevertheless notable divides in Internet penetration still exist between Americans with different levels of income and education, different racial and ethnic groups, old

and young, single and dual-parent families, and those with and without disabilities.[32] Many industry leaders in the corporate sector have expressed concern that too many people are being left behind in the Information Age, and multiple nonprofit organizations and foundations have highlighted this problem.[33] Governments in Finland, Germany, Canada, and Sweden have all announced programs to address access inequalities, often blending private and public resources. The British government, for example, has established a network of city learning centers, introduced a scheme to distribute reconditioned computers to homes in poor neighborhoods, and developed a national grid linking all public libraries to the Internet.[34]

Will digital inequalities prove a temporary problem that will gradually fade over time, as Internet connectivity spreads and "normalizes," or will this prove an enduring pattern generating a persistent division between info-haves and have-nots? Again the debate divides cyber-pessimists who emphasize deep-seated patterns of social stratification and the growth of an unskilled underclass in technological access, cyber-skeptics who believe that technologies adapt to society, not vice versa, and cyber-optimists who hope that in affluent postindustrial societies, at least, the digital divide will eventually succumb to the combined forces of technological innovations, markets, and the state. Positive scenarios suggest that inequalities in Internet access may prove a short-term phenomenon, similar to the type of households that could afford to buy television sets when services were first introduced in the early 1950s. In this perspective, the profile of the online community will probably come to reflect society as a whole given the wider availability of simpler and cheaper plug-and-play technologies and faster broadband services, facilitating delivery of popular mass entertainment including streaming video-on-demand. Some suggest that high-tech companies will compete to connect the public with a speed and efficiency that no government program can match, even in the neighborhoods of the urban poor, if there is mass demand for the services.[35] For those with personal computers, free Internet services, email and Web hosting services are already widely available, albeit with advertising strings attached.[36] The market may be insufficient to close the gap but the nonprofit sector has also been active. Major American corporations including Microsoft, Intel, Hewlett-Packard, and AT&T have foundations devoted to expanding access to local communities, most often through donating educa-

tional equipment and fostering training in deprived areas, complementing state initiatives designed to furnish the younger generation with keyboard skills and training in wired schools. Telecommunications policy may play an important role here if the Internet is treated as a public utility, so that access is made widely available through public libraries, community centers, and private homes, much as telephone services were regulated to produce low-cost services and universal access to rural areas.[37]

The interesting question is not whether there will be *absolute* social inequalities in Internet access; of course there will be, as in other dimensions of life. Although Alexander Graham Bell's commercial telephone service was launched in the United States in 1877, today in America, more than a century later, there remain pockets of racial inequality in access to household telephones. Cable TV started to become available in the mid-1960s but today, owing to choice or necessity, only two-thirds of American households are connected, along with about half of all households in industrialized nations.[38] Given substantial inequalities in the old mass media, it would be naive to expect that the Internet will magically transcend information poverty overnight. The more intriguing series of questions addressed by this book concern whether there are special barriers to digital technologies, such as their greater complexity or costs, and whether *relative* inequalities in Internet use will be similar to disparities in the penetration rates of older communication technologies.

THE DEMOCRATIC DIVIDE

The last challenge, and perhaps the most intractable, concerns the potential impact of the digital world on the distribution of power and influence in political systems. Even if we assume, for the sake of argument, that Internet penetration rates will gradually widen throughout society there is growing awareness that a substantial *democratic divide* may still exist between those who do and do not use the multiple political resources available on the Internet for civic engagement. What will be the impact of digital technologies in the public sphere?

The Internet has generated deeply contested alternative visions about the future. Cyber-optimists emphasize the Panglossian possibilities of the Internet for the involvement of ordinary citizens in direct democracy. Digital technologies hold promise as a mechanism facilitating alternative channels of civic engagement such as political chat rooms, electronic voting in general elections and for referenda issues, and the mobilization of virtual communities, revitalizing levels of mass participation in public affairs.[39] The use of the Internet by groups and social movements is often believed to exemplify digital politics. This view was popular in the mid-1990s and the revolutionary potential of digital technologies continues to be expressed by many enthusiasts such as George Gilder.[40] Yet as the Internet evolved, a darker vision has been articulated among cyber-pessimists who regard digital technology as a Pandora's box unleashing new inequalities of power and wealth, reinforcing deeper divisions between the information rich and poor, the tuned-in and the tuned-out, the activists and the disengaged. This account stresses that the global and social divides already discussed mean that Internet politics will disproportionately benefit the elite.[41] In this perspective, despite the potential for technological innovations, traditional interests and established authorities have the capacity to reassert their control in the virtual political sphere, just as traditional multinational corporations have the ability to reestablish their predominance in the world of e-commerce.[42] Finally, cyber-skeptics argue that both these visions are exaggerated, because so far the potential of the Internet has not had a dramatic impact on the practical reality of "politics as usual," for good or ill, even in countries at the forefront of digital technologies.[43] For example, during the 2000 American presidential campaign the major candidates used their Web pages essentially as glossy shop-windows, as fund-raising tools, and as campaign ads, rather than as interactive "bottom-up" formats for public comment and discussion.[44] Technology, in this view, is a plastic medium that flows into and adapts to preexisting social molds. The demise of many dot.coms in the business world has reinforced the skeptical view.

Each of these viewpoints reflects an element of truth depending, like a Rorschach test, on whether studies are focusing on different multifaceted components of digital technologies. As with the blind men of Indostan in Hindu legend, observers touch different parts of an elephant – the tusks, the tail, and the trunk – and report their experience with absolute conviction as though describing the whole of the digital world.[45] Yet it requires a considerable stretch to get our arms around this beast. Multiple warnings should be posted before entering this territory. Deep-rooted hopes and fears about the poten-

tial for technology often outweigh dispassionate analysis. Powerful myths and vivid anecdotes commonly appear as plausible as concrete observations. The best forecasts often seem little more than intelligent guesses. "Facts" commonly exhibit a shelf life of weeks or months. And hucksters in the guise of market research hype the industry's wares.

THE CORE ARGUMENT, CONCEPTUAL FRAMEWORK, AND STRUCTURE OF THE BOOK

How can we move beyond speculative theorizing toward more systematic evidence on these issues? The overall structure of the book can be summarized as follows. Chapter 2 considers approaches to understanding the causes and consequences of Internet access and use, and the major challenges that arise owing to the rapid pace of technological and social development, the limitations of cross-national comparative evidence, and the need for a multimethod research design. The chapter concludes that the most effective way to meet these challenges is to develop a comparative multilevel research design covering a wide range of political system.

In this study, the conceptual framework used to understand these issues distinguishes among three nested levels of analysis, as illustrated in Figure 1.1. The *national* context, including the macrolevel technological, socioeconomic, and political environment, determines the diffusion of the Internet within each country. The *institutional* context of the virtual political system provides the structure of opportunities mediating between citizens and the state, including the use of digital information and communication technologies by governments and civic society. Finally, the *individual* or microlevel of resources and motivation determines who participates within the virtual political system. Most studies are limited to only one level. In contrast the more holistic approach used in this book compares the national context of Internet access in 179 countries around the globe, as well as the virtual political system within these nations, and then explores patterns of online civic engagement among individual citizens in Western Europe and the United States. The nested framework assumes that the national context, such as the process of technological diffusion, influences the development of the virtual political system. In turn, the core institutions of the political system available in the digital world provide the

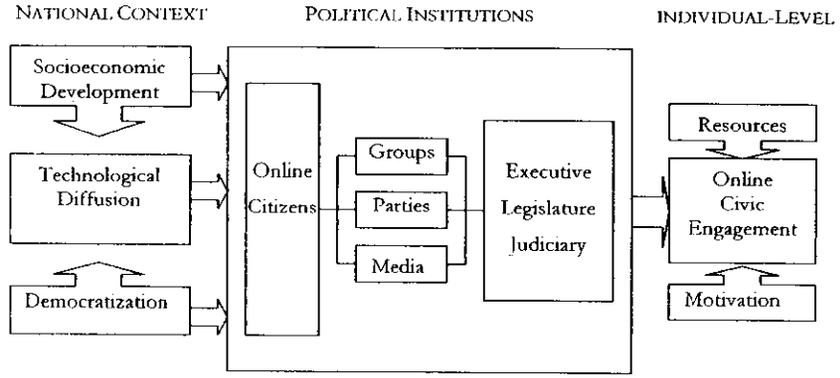


Figure 1.1. The Internet Engagement Model.

systematic context within which individual citizens have opportunities to participate online. Which particular citizens choose to take advantage of these opportunities is determined by their personal resources (like time, money, and skills) and their motivation (like interest, confidence, and efficacy).

Those preferring to go directly to the meat and potatoes of the evidence can turn directly to Chapter 3 which establishes what we know about the global divide in the networked world, drawing upon aggregate indicators to map the spread of digital technology around the globe, and then considers the causes of cross-national differences in Internet connectivity. The evidence indicates that some developing nations such as Malaysia, Brazil, and Taiwan have made substantial progress in the knowledge economy. But average rates of Internet penetration have grown sluggishly, at best, in most developing nations. Chapter 3 demonstrates that the global divide in Internet access is substantial and expanding: About 87 percent of people online live in postindustrial societies.[46] The contrasts worldwide are sharp: More than half of all Americans now surf the Internet compared with 0.1 percent of Nigerians. There are currently twice as many users in Sweden than across the vast continent of Sub-Saharan Africa.[47] In considering alternative explanations of this phenomenon, the evidence strongly suggests that economic development is the main factor driving access to digital technologies, so that the Internet reflects and reinforces traditional inequalities between rich and poor societies. Once we control for levels of economic development, then democrati-

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zation plays an insignificant role in the process of technological diffusion. Far from a new pattern, the global spread of the Internet reflects existing patterns of access to the traditional mass media including television, newspapers, and radios, disparities that have existed for decades and that show no sign of gradually closing over time. Striking inequalities are evident worldwide: Half a billion people living in Sub-Saharan Africa share 14 million phone lines, fewer than in Manhattan or in Tokyo.[48] In Sub-Saharan Africa, for every 100 people there are only 17 radio sets, 5 televisions, and 0.5 percent mobile phones.[49] On this basis it seems likely that, despite initiatives by state and international agencies, and despite technological developments in the marketplace, the global digital divide will probably continue in the foreseeable future, driven by world poverty, even if new forms of Internet transmission eventually become inexpensive and as easy as pushing the power button on a radio.

Chapter 4 goes on to analyze the extent and the causes of social inequalities in digital opportunities within different countries, focusing on Internet penetration rates broken down by social class, education, gender, and generation. The composition of the online population is analyzed using representative surveys in Western Europe and the United States. The study concludes that unequal rates of Internet penetration are due to deep divisions of social stratification within postindustrial societies – such as patterns of household income, education, and occupational status – that shape not just digital opportunities but also access to other common forms of mass communications including cable and satellite television, VCRs, and fax machines. Far from narrowing as the information society expands, the income gap in Internet penetration is currently greatest in societies such as Sweden and the Netherlands where access to digital technologies has become most widespread. Of course considerable caution is needed in projecting from current patterns to future trends. The rosy scenario suggests that digital opportunities could eventually become more socially inclusive under certain conditions: if costs continue to fall dramatically in the marketplace, if the technology becomes simplified, and if policy initiatives by the state widen Internet access, training, and keyboard skills. Through inexpensive cell phones or handheld personal assistants, use of a stripped-down version of the Internet, for example just email and some headline news services could eventually become as ubiquitous in postindustrial societies as the availability of household television sets. Genre-scrambling

technologies converging broadband access, the Internet, telephony, and TV entertainment promise to alter conventional forms of content delivery and also inputting devices. The long-term process of generational replacement should eventually lead to greater familiarity with computers throughout society.

But in the short-term these rosy projections, while not impossible, involve multiple "ifs." At present, affluent households with multiple consumer durables designed for traditional forms of home entertainment and communications are also most likely to possess networked personal computers. Poorer families are excluded from digital opportunities, and hence access to online employment vacancies, educational resources, and social networks. Moreover, even if basic access to email becomes ubiquitous, say as common as public telephones in Europe and North America, the marketplace for technological innovations will continue to generate ever faster, smaller, and better machines, spawning new applications and multiple levels of functionality. The chameleon-like capacity of digital technologies to morph, converge, and reappear in different guises, as cell phones can play music files, personal digital assistants can take photos, and computers carry radio waves, makes the Internet dissimilar to earlier machines like television sets. Even if the basic digital divide shrinks gradually over time, it is naive to believe that the virtual world can overturn fundamental inequalities of social stratification that are endemic throughout postindustrial societies, any more than it is likely to overcome world poverty.

THE VIRTUAL POLITICAL SYSTEM

Part II compares the institutional context for representative democracy focusing on three issues: Where and what type of political organizations worldwide have adapted to digital technologies? What are the functions of these websites for maximizing transparent information and interactive communications? And what explains the rise of digital politics, in particular the relative significance of socioeconomic development, technological diffusion, and the process of democratization? Chapter 5 expands upon theories of cyber-democracy and considers the potential capacity of the Internet for strengthening civic society and the institutions of representative democracy around the world. Although many

specific case studies describing cyber-politics in particular nations are becoming available, and a burgeoning literature is developing in the United States and Western Europe, it remains difficult to find systematic typologies and evidence comparing digital politics across a wide range of countries at different levels of social, economic, and political development. Subsequent chapters compare the way that the institutions of representative democracy have responded to digital politics, drawing on evidence about the distribution and function of websites for different types of political organizations from around the world. Chapters 6 through 9 analyze which countries have forged ahead in digital politics, where the Internet has been used for information and communication by governments and civic society, and the socioeconomic, technological, and political factors driving the adaptation of organizations to digital politics. As noted earlier, there are many reasons to be cautious in any analysis. The first decade of the emerging Internet age has seen a process of restructuring and adaptation as political institutions have learned what does, and doesn't, work using digital technologies. Yet precisely because this is a period of experimental transition and institutional change it is particularly important to draw the appropriate lessons based on the available evidence, to map the current state of play, and to consider how the Internet functions in a wide range of political systems, including but also beyond the United States and Western Europe.

The optimistic claims that the interactive capacities of digital technologies will facilitate a new era of direct democracy, characterized by widespread citizen deliberation in affairs of state, like a virtual Agora, while attractive as a normative ideal, is ultimately implausible in practice as soon as we understand who becomes involved in digital politics. As we will see, the cross-national survey evidence indicates that those who take advantage of the opportunities for electronic civic engagement are activists most likely to participate via conventional channels. As a medium of choice par excellence, it seems improbable that digital politics will reach the disengaged, the apathetic, and the uninterested, if they choose to spend their time and energies on multiple alternative sites devoted to everything from the stock market to games and music. In this regard, the Internet seems analogous to the segmented magazine market, where some subscribe to *The Atlantic Monthly, The Economist,* and *Foreign Affairs,* but others pick *Golfing Weekly* or *Playboy.* The available studies of politically oriented discussion groups, bulletin boards, and online chat rooms have found these largely fail as deliberative fora,

instead serving as places to reinforce like-minded voices.[50] Claims for the potential of digital direct democracy to revitalize mass participation can find few crumbs of support from these studies. At the same time, the skeptics' claim that nothing much will change in the political system, as most established political institutions will adapt digital technologies to facilitate existing functions, while admittedly more realistic, overlooks the occasional indications that, here and there, now and then, like a faint sporadic seismic tremor, some disruptive threats to politics as usual are already becoming evident.

Rejecting the view that either everything will change as direct democracy comes to replace representative governance, or that nothing will change as the digital world merely replicates "politics as usual," this book argues that digital technologies have the capacity to strengthen the institutions of civic society mediating between citizens and the state. Established political institutions, just like major corporations, can be expected to adapt the Internet to their usual forms of communication, providing information online, but not reinventing themselves or rethinking their core strategy in the digital world, unless successfully challenged. In contrast, insurgent organizations traditionally have fewer political assets, fewer traditional advantages, but also fewer inhibitions about adapting flexibly to the opportunities for information and communication via the Internet. If this account is essentially correct, digital politics may have most impact in leveling the playing field, not completely but at least partially, for a diverse range of challengers, such as transnational advocacy networks, alternative social movements, protest organizations and minor parties, such as those concerned with environmentalism, globalization, human rights, world trade, conflict resolution, and single-issue causes from all shades of the political spectrum, ranging from genetically modified food and anti-fuel taxes to animal rights and anti-sweat shops. The Internet does not drive these movements – these causes are triggered by deeper passions – but it facilitates their organization, mobilization, and expression.[51]

Information and the mechanisms for delivering it are the lifeblood and sinews of the body politic. Some power comes out of the barrel of a gun. Some power can be bought with the resources of wealth and income. Some may be inherited by sultans and princelings. But in democratic systems the primary coinage of the realm – the resource that persuades, that influences, that swings votes – is information. "Information" comes in all shapes and forms, from the publication of

official documents by government departments to brief news bulletins on the hour, from lengthy parliamentary debates to 30-second campaign ads, and from demonstrations by new social movements to informal conversations over the water cooler. Political organizations are essentially designed as control systems for the transmission of information, binding together the activities of all members within the unit and communicating priorities to the external world. Some information exchanges are brief and transitory; others use rich and well-developed channels. The explosive growth of connectivity via the Internet alters the transmission of information among networks, shrinking costs, maximizing speed, broadening reach, and eradicating distance. Potentially these changes can have profound consequences for altering the balance of resources and power between outsider challengers and established organizations within the political system. Hierarchical communication channels, typical in bureaucratic organizations like government departments and international agencies, are less effective and slower mechanisms of information transmission than horizontal networks shared by informal coalitions of alternative social movements. National boundaries to information flows dissolve, allowing global networks to flourish. Independent upstarts and multiple sources of "news," where immediacy outweighs authority, threaten the legitimacy of traditional journalism in the newspapers and television. Communication costs fall, and information costs plummet even faster. With wider and easier access to official sources, opposition groups and social movements can challenge the authority and expertise of government ministers, civil servants, and elected officials on their own turf.

The main democratic potential of digital information and communication technologies lies in strengthening organizational linkages and networking capacities in civic society. Strengthening these bonds, it will be argued, has the capacity to produce sudden disruptions to politics as usual, especially for flash coalitions mobilizing suddenly like a guerrilla army then dissolving again, exemplified by events such as the anticapitalism violent protest in the City of London in June 1999, direct-action campaigns against the World Trade Organization on the streets of Seattle and Quebec, antiglobalization protests against the World Bank/International Monetary Fund in Prague and Washington, D.C., and the poujadist fuel price revolt by farmers and truckers that swept the European continent in October 2000. Such occurrences remain relatively rare, but they can have immediate impact on the policy process, and they are important as indicators of the disruptive potential of digital politics. Some flash protests are temporary phenomenon. Other transnational advocacy networks manage to sustain longer-term electronic coalitions, such as the International Campaign to Ban Landmines that resulted in a treaty signed by 122 nations in 1997. Global protest movements and direct-action demonstrations spreading across national borders have existed for decades, such as the antinuclear movement in the 1950s and the anti-Vietnam war protests of the 1960s, or even farther back the antislavery and the suffrage movements in the nineteenth century. The phenomenon is far from new but these movements are facilitated in an environment of minimal-cost instantaneous global communications. Governments, like British redcoats lined up in perfect formations, seem unsure how to respond. They are flustered when suddenly outmaneuvered by the ad hoc coalitions of truck drivers and fuel-tax protestors, the environmental activists and animal-rights lobbies, the anticapitalists and antiglobalist forces. It is true, as cyber-skeptics claim, that most established political institutions prefer to co-opt the capacities of new technologies to preexisting functions, rather than being forced to reinvent themselves in the Internet age. But it is also true that the capacities of the Internet are adapted more easily by smaller, more flexible organizations, a process that is particularly important for the process of democratic consolidation, and for opposition movements seeking to challenge authoritarian rule around the globe.

THE IMPACT ON CIVIC ENGAGEMENT

What will be the impact of this process for civic engagement among ordinary citizens? Part III goes on to examine the nature of the cyberculture and the influence of digital politics on public participation, and then summarizes the core thesis argued in this book. Chapter 10 analyzes political attitudes in the United States and Western Europe. Many have concluded that as the Internet population has gradually normalized in America, the digital world has come to reflect the general population.[53] Nevertheless a more detailed examination of the values and attitudes of the online community in America and Europe, where we have survey evidence, suggests the existence of a distinctive cyberculture, one favorable toward the "new" left on the social agenda and the "old" right on the economic dimension. Just as Internet enthusiasts sympathize with nonregulation in the sphere of personal lifestyles, so they favor freedom from government in the economic sphere. Moreover, this cyberculture is not simply a by-product of the social profile of those who go online,

since this pattern remains distinctive even after controlling for the usual demographic factors such as the age, education, sex, and income of the online population. Such a cyberculture is one broadly sympathetic to the alternative social movements that use digital technology most effectively for direct action and protest demonstrations.

Will the Internet have the capacity to revitalize public participation in conventional politics, such as levels of party membership, electoral turnout, or activism in civic and voluntary organizations? Chapter 11 suggests that digital politics reduces some of the information and communication costs for individual citizens who are interested in public affairs, but at the same time the Internet probably has the least impact on changing the motivational basis for political activism. In this way, digital politics functions mainly to engage the engaged. For those with access and motivation, the Internet facilitates opportunities for civic engagement, increasing the ability to drill down and compare multiple news sources on an issue, to forward articles and clippings to colleagues, friends, and family, to donate funds electronically to causes or election campaigns, to support groups mobilizing around particular issues, to organize within local neighborhoods, and to discuss politics online, as well as to research official documents and legislative proposals, to access government services and download official forms, and to contact public servants about particular problems of health or housing. Reduced information and communication costs lower some, although not all, of the barriers to civic engagement. Costs can only be expected to fall with the expansion of online political resources, giving grounds for optimism about the ability of digital politics to revive activism among the active. Yet the evidence also suggests that, at least in the short term, at individual level, altering the structure of opportunities and the balance of relevant resources probably has minimal impact on changing the motivational basis of political participation and interest among the mass public. Digital politics thereby contributes toward the vitality of representative democracy, but it also largely bypasses the disengaged.

In this regard, the role of the Internet is similar to the impact of traditional forms of mass media. Previous work has established a consistently positive association between use of the news media and indicators of civic engagement in the United States and Europe.[54] Those who watch the news and current affairs on television, read newspapers, and listen to radio news were found to be more politically informed, trusting, and active than average, even with the usual controls for social

background such as age, gender, education, and income. The evidence in this book confirms that, along similar lines, those already most interested and involved in public affairs take most advantage of the new opportunities for information, expression, and political mobilization available via the Web. Environmentalists, for example, are most likely to surf the Greenpeace website, just as Republicans are most likely to check *www.Bush2000.org,* and women are most likely to click on *www.Oxygen.com.* Like discussing gun control or abortion over dinner with like-minded friends, reading liberal op-ed pages on problems of health care or affirmative action in schools, or attending a protest rally about genetically modified food, this experience can be expected gradually to *reinforce* political attitudes and strengthen the involvement of the participants. This process remains important, functioning to encourage the involvement of ordinary citizens in democratic government through representative channels. Yet it disappoints those who hope that the Internet will function as a deliberative public forum, drawing the less engaged into civic life, replacing representative institutions, and thereby strengthening direct, plebiscitory, or "strong" democracy.

Therefore the theory developed in this book attempts to strike a balance between more pessimistic claims that the development of the Internet will serve to reinforce the voices of the powerful, the more skeptical claims that it will merely reflect "politics as usual," and the more optimistic claims that cyber-democracy will transform governance as we know it and restore levels of mass political participation. Instead, the book concludes that the restructured opportunities for information and communication available via digital politics will potentially have positive consequences for civic society, altering the balance of relevant resources and slightly leveling the playing field. The primary beneficiaries are likely to be marginal groups such as minor and fringe parties, loose coalitions of protest organizations, and alternative social movements, particularly those advocating causes that are most conducive to the cyber-culture. Reducing the costs of information and communication minimizes some, although not all, of the significant barriers to effective political participation at individual level; it becomes easier for ordinary citizens to learn about public affairs, if they are so inclined, and to express their views and to mobilize. This process is most important in many consolidating democracies, stranded midway between an authoritarian past and stable democratic future. The wider diffusion of digital technologies can play a significant role in

strengthening civic society in countries such as Taiwan, Brazil, and South Africa if e-governance improves transparency and openness in the policymaking process, if parliaments and parties use new media to strengthen their internal organizations and their links with the public, and if opposition movements develop virtual coalitions to challenge the predominance of the government's message in television, radio, and newspapers. But whether the Internet can ever encourage the less engaged to take advantage of these opportunities at mass level remains doubtful, because as the medium of choice par excellence, it becomes even easier for people to tune out from public life.

Of course, as discussed in the next chapter, there are strong grounds for caution in any prognostication about future developments. This discussion relates to the use of digital technologies during the first decade of the emerging Internet age, and the long-term consequences of these developments cannot be predicted with any accuracy at this stage. History furnishes numerous examples of the failure to foresee the ultimate uses of technologies at the time when they were first introduced. Newfangled telephones were first thought of as channels of musical entertainment, not personal communications. In the nineteenth century, modest electric shocks were believed the novel cure perfect for improving the healthy constitution. When wireless amateurs started broadcasting before World War I, most saw radio as an active medium of communication, a hobby for young boys, not a passive listening experience. Forecasts often fail to predict the weather, the election results, or the stock market for the day after tomorrow, let alone for decades from now. Contemporary estimates for the impact of the Internet may be similarly misplaced. Digital politics has evolved rapidly during the last decade, and multiple developments will probably occur within the next, such as online registration and voting. The long-term impact of digital technologies could ultimately produce different consequences to their effects during the emergent era. But despite the importance of considerable caution, the pattern of global, social, and democratic inequalities described in this study fits what we already know about the impact of traditional forms of political communications, like newspapers, radio, and television, and also receives support from the comparative evidence in the emergent Internet age, so that the evidence deserves to be examined with an open mind to contrary data and countervailing indicators. By systematically comparing the diffusion of digital politics around the world, including both the leaders and

laggard nations, this account can be tested to see whether it provides useful insights into the spread of the Information Society in recent years. In conclusion, Chapter 12 recapitulates and expands on the core theory at the heart of this book, summarizes the evidence for this interpretation, and considers the broader implications for understanding digital politics in the Internet age.

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CHAPTER 2

Understanding the Digital Divide

Debate about the impact of the rise of the Information Society has produced deeply contested visions predicting the future direction of trends. Optimists hope that the development of the Internet has the capacity to reduce, although not wholly eradicate, traditional inequalities between information-rich and -poor both between, and within, societies. In contrast, pessimists believe that the digital technologies will reinforce and exacerbate existing disparities. Skeptics suggest that both the fears and hopes are exaggerated, with technologies adapting to the social and political status quo, rather than vice versa. What evidence would help to settle these claims? How can we move from the Frank Capra and the Ingmar Bergman visions toward a more systematic understanding of the impact of the Information Society? It remains difficult to sort the facts from the hype, despite the burgeoning literature on all aspects of the Internet ranging from Web design, software development, and e-commerce to the sociology of the network society, group identities, and virtual culture. Studies in any discipline assessing the impact of the Internet face three main challenges: the problems of studying a phenomenon undergoing rapid change; the limitations of the available cross-cultural evidence allowing us to generalize beyond the experience of the United States; and the difficulties of developing and integrating triangulated methodologies drawn from different disciplines.

THE RAPID PACE OF CHANGE

The first challenge is the rapid pace of technological innovation and social adaptation so that studies of the impact of info-tech represent blurred snapshots of a moving bullet.[1] The genesis of the Internet was

initially fairly slow but postindustrial societies are currently experiencing a sharply accelerating 'S' curve of diffusion. The birth of computer-mediated networked communication can be traced back to ARPANET in 1969, an experimental four-computer network, established by the Advanced Research Projects Agency (ARPA) of the U.S. Defense Department to develop a secure form of communication via multiple destinations in the event of nuclear war. Information was split up into "packets" that were then transmitted via several dispersed routes: if one link was unavailable then, like a delta river with numerous tributaries, information simply traveled through alternative routes before the packets were reassembled at the destination. In the 1970s, dispersed communication networks spread email among a select community of scientists and scholars at elite universities and research centers. In 1971 ARPANET linked about two-dozen computers ("hosts") at 15 sites, including MIT and Harvard, and a decade later more than 312 hosts were networked. This process was accelerated in 1986 by the National Science Foundation's development of a high-speed backbone network to link science and engineering, although, other than email, this remained mainly the domain of computer-science aficionados happy to struggle with unforgiving lines of computer programming and printouts. Data were still routinely delivered on magnetic tape mailed in large tin cans like movie reels. Beyond linking communications between research institutions and scholars, the most popular uses of the new networks were financial transactions in electronic banking and email for business.

The Internet as we know it today came about with the invention in 1989 of the World Wide Web and a hyper-text language for global information sharing, by Tim Berners-Lee at CERN in Geneva, and the subsequent release in 1991 of the first client browser software for accessing materials on the Internet. At this time about twenty countries were connected to the network, mostly in North America and Western Europe. The decisive technological breakthrough popularizing the medium occurred in 1993, when the National Center for Supercomputing Applications released Mosaic, the first graphical Web browser, made available for Unix systems, then for Microsoft Windows and the Apple Macintosh. The graphical browser removed the need for any technical expertise in accessing the Web beyond the ability to point and click, making it instantly accessible to a five-year-old. The remarkable rise of the Internet as a new mass medium came in October 1994 when Netscape Communications released the Netscape Navigator browser, built on Mosaic technology and distributed free. Microsoft awoke rela-

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tively late to the opportunities of the Internet but eleven months later, in August 1995, Internet Explorer was released, bundled with the launch of Windows 95.

In postindustrial societies, the Internet wildfire during the last decade has been, as everyone observes, remarkable. The earliest estimates suggest that in 1994 there were about 3 million users worldwide, mostly living in the United States.[2] The following year this number had risen to 26 million. The online population has subsequently roughly doubled every year since then, reaching an estimated 407 million people by late-2000 (see Figure 2.1). The first-ever American opinion poll on this issue, conducted by Louis Harris and Associates, found that one-third of the public had heard of the Internet in June 1994 but only 7 percent had ever used it.[3] Pew surveys estimate that the following year the proportion of users had doubled to about 14 percent of all Americans, but by mid-2000 more than one-half of all Americans used the Internet (54 percent).[4]

Therefore as a form of information and communications spreading beyond the scientific and technical elite, the Internet as a mass medium remains a relatively recent development. Computers have been around for about fifty years, and distributed computer networks for about thirty years, but the popular point-and-click World Wide Web, as we know it, has only existed since 1993. Predictions suggest that the familiar Internet experience of the first decade – with email and Web pages delivered through wired umbilical cords to beige desktop boxes – will probably not be the familiar Internet experience envisaged for the next decade, with at least a cut-down version enabled through wireless cell phones like DoCoMo services in Japan, pagers, digital televisions, handheld personal assistants like Palm Pilots, even streamed in headline versions through ATM banners and screens fitted in elevators, bus stops, and airports, with online automobiles so that we are All Internet, All the Time. For technophiles, the Web is promised to arrive through everything from our toasters to our televisions. Although it is difficult to sort out the reliable estimates from the industry hype, market research forecasts suggest that by 2005 more than one-half of all Americans online, and almost three-quarters of worldwide users, may have digital Web appliances to download information.[5] Novel "killer apps" are predicted to transform information technologies.[6] Yet predictions are in constant danger of being overtaken by events ("So 1998."), as well as being exaggerated into hyperbole by the industry in its own interests. Despite predictions that smaller firms will thrive in the new economy, multiple

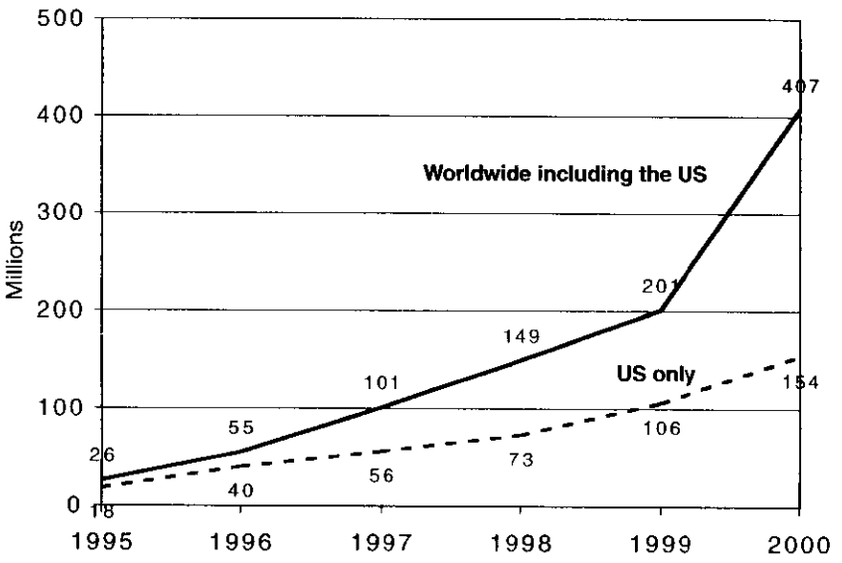


Figure 2.1. Worldwide Trends in the Online Population, 1995–2000.
Source: "How Many Online", *www.NUA.ie,* November 2000.

entrepreneurial Internet start-ups have fallen by the way, while corporate mergers producing multinational companies are more fashionable today than ever before. Like the Nasdaq, past irrational exuberance surrounding digital technology stocks may prove an unreliable guide to future performance.

One of the best ways to understand the rapid pace of change is to monitor trends during the last decade across many different nations at the forefront of the emergent information society, in order to understand how digital politics has evolved in response to the new structure of opportunities. The past decade is likely to prove atypical of subsequent developments as people learn what does, and doesn't, work. In the 1998 U.S. elections, for example, only one in ten of the major party Senate, House, and gubernatorial candidate websites facilitated online campaign donations and almost one-half did not even ask for money.[7] In contrast, just two years later John McCain's campaign raised $1 million via online contributions in the 48 hours after his New Hampshire victory in February 2000, or $2.5 million in total online.[8] In 1996, grassroots activists and individual voters for or against the major presidential candidates in America set up a handful of homegrown sites. In 2000, in contrast, there were almost 7,000 such sites. In 1996,

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just a handful of dedicated Internet news outlets got to the U.S. presidential party conventions. Four years later there were 80 to 100 such outlets, like *Slate* and *Salon*, as well as online coverage by almost every traditional news organization including CNN and C-Span. The key issue with these sorts of developments is how the public responds when digital politics evolves. When government departments go online, how do people use these sites to seek information? When parties, groups, and campaigns use horizontal networks via "virtual" conferences, policy discussions, and innovative feedback mechanisms, does this mobilize supporters? What new formats work, and what don't? Although future developments remain uncertain, the 1990s represents a unique opportunity to capture how the first generation of online users evolved, similar to studies in the 1950s analyzing the early television audience.

One way to think about these issues is to draw upon classic theories of technological diffusion developed by the work of the nineteenth-century French sociologist Gabriel Tarde and by the Harvard sociologist Pitirim Sorokin (1941) and advanced by communications scholars Elihu Katz and Everett Rogers.[9] These theories suggest that the adoption of many successful innovations – whether of new strains of seed corn, industrial machinery, or new medical breakthroughs – have commonly followed an S-(Sigmoid) shaped pattern.[10] New technologies have often experienced a slow rate of initial adoption, followed by a substantial surge that peaks when penetration levels reach saturation point and demand subsequently slows. Cyber-optimists suggest that the spread of the Internet will follow a *normalization* pattern, as costs fall, as the technology becomes simplified allowing plug-and-play access, and as the Web increasingly provides mass entertainment and cheap communications via streaming audio and video. In the normalization model illustrated in Figure 2.2, those who adopt the innovations at an early stage will be ahead of the curve, with the resources, skills, and knowledge to take advantage of digital technologies, but in the long term cyber-optimists believe that penetration will become saturated in these societies. Once a high proportion of households have a personal computer and access to the Internet – like owning a refrigerator, automobile, or washing machine – then demand will slow. The theory predicts that given saturated demand, prices will fall further to attract new users, allowing laggards to catch up, so that eventually access to digital technologies becomes pervasive. The initial period of adoption may therefore be expected to widen social inequalities but the normalization hypothesis suggests

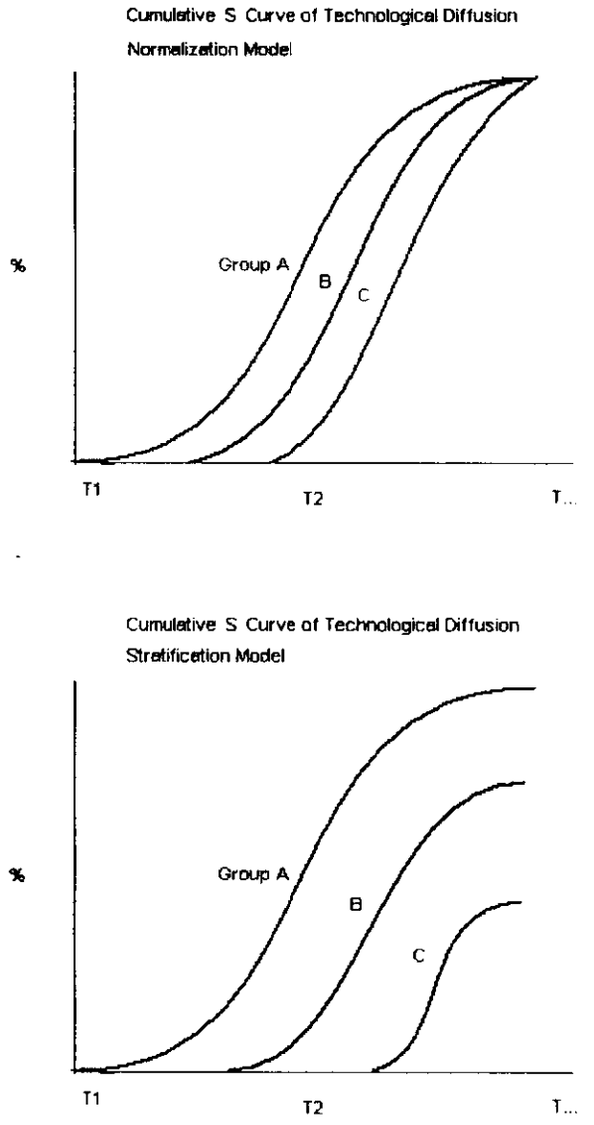


Figure 2.2. The Cumulative S Curve of Technological Diffusion.

that this temporary gap will eventually close. In contrast, cyber-pessimists emphasize that the *stratification* model provides a more realistic scenario where groups already well networked via traditional forms of information and communication technologies will maintain their edge in the digital economy.

Diffusion theory allows us to compare the growth of personal computers and the Internet with earlier technologies. In the United States, the spread of many previous innovations has usually followed a sigmoid (S-shaped) time path characterized by a slow pace of initial adoption, followed by a significant advance, and then a gradually tapering of demand (see Figure 2.3). Televisions in America experienced a rapid surge of sales in the 1950s, fueled by pent-up demand for consumer goods and the hiatus in TV production and broadcasting during World War II. VCR sales saw a similar surge in America during the late 1980s. In contrast, some other communication technologies took far longer to spread throughout the American population. Sales of radio receivers were initially held back by the technological complexity of crystal sets and the onset of the Great Depression in the 1920s, before experiencing a slow and steady rise, until today there are more radio sets than people in the United States. The telephone, which had been available as a commercial service since 1877, only took off for the majority of American households after World War II. Automobiles, as big-ticket household items, also experienced a steady climb in sales after 1945 until reaching a plateau in the 1980s. Cable TV saw slow diffusion in America from 1960 to 1980, due to the investment costs of laying cable and the number of stations available to most subscribers, before accelerating in availability. In the United States, the flood of Internet users since the early 1990s has followed an S-shaped curve, and it remains to be seen whether this curve will bottom out with two-thirds access, like cable TV, or more than 90 percent access, like TV. The pattern of American adoption so far has been closer to the rapid surge in television sets and VCRs rather than the slower diffusion of telephones and radios. Although the Internet remains a relatively new phenomenon, the diffusion patterns evident in related communication and information technologies in America provide important evidence about what we might expect to occur with the growth of the online community in future decades and in other postindustrial societies at the forefront of the knowledge economy, such as Sweden and Australia.

Worldwide the comparison of the spread of radios and televisions since the 1950s, and the rise of the Internet since 1995, shows more gradual secular trends rather than a sudden S curve (see Figure 2.4). The growth of the online community has been substantial: For comparison with previous innovations, the telephone took close to 75 years to reach 50 million users worldwide, and television took 13 years, but it took only 4 years for the Internet to reach the same number.[11]

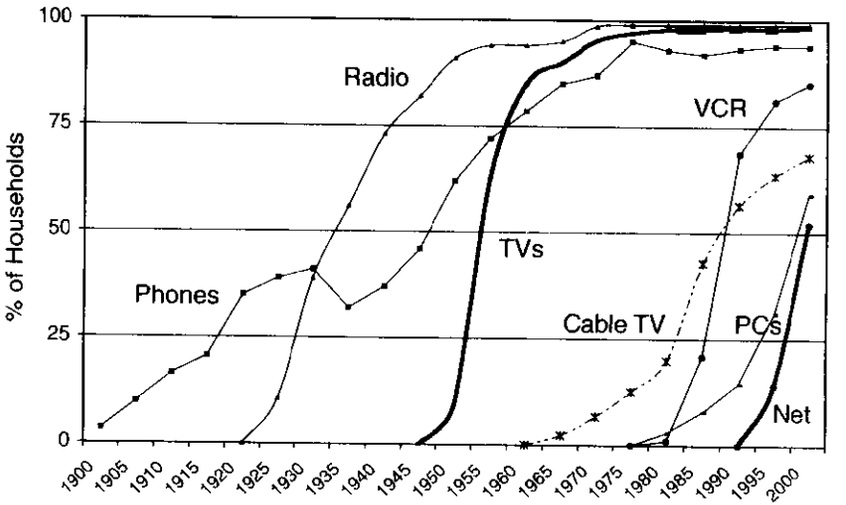


Figure 2.3. Twentieth-Century American Technology.
Sources: U.S. Census Bureau: *Statistical Abstract of the U.S.,1999; Historical Statistics of the U.S.*

CROSS-NATIONAL EVIDENCE

In addition to drawing comparisons historically over time, another major challenge concerns the difficulties of generalizing across many countries based on the limited evidence. Most studies of digital politics focus on the United States yet these findings may well be, in this, as in so much else, exceptional.[12] As an industry leader, the United States is certainly atypical in Internet use, even within the universe of postindustrial societies – containing an estimated three-fourths of all e-commerce sites worldwide, 79 percent of the world's Internet hosts, 59 percent of the world's electronic mailboxes, 54 percent of online buyers, and 38 percent of Internet users.[13] If access to digital technologies is heavily contextual, depending on the structure of opportunities available within each society, then the typical experience of Silicon Valley dot-com entrepreneurs, Harvard undergraduates, and New York lawyers will probably have little in common with their counterparts in London, Paris, and Tokyo, still less in Moscow, Beijing, and Johannesburg. A broader analysis, which examines global patterns, contrasting leaders and laggard societies, provides the basis for more reliable generalizations.

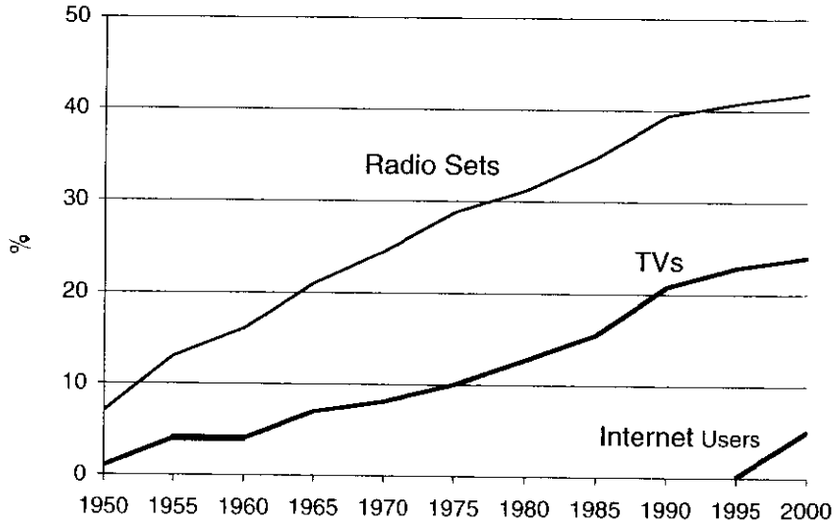


Figure 2.4. Worldwide Diffusion of Radio, Television, and the Internet, 1950–2000.

Sources: **Television sets and radio receivers per household:** annual UNESCO *Statistical Yearbooks*, UNESCO, Paris; **estimates of the Internet population:** "How Many Online?", *www.NUA.ie.*

There is another important reason why we need comparisons beyond the United States. Democracies differ significantly in their core institutions and constitutional features, most notably in terms of majoritarian or proportional electoral systems, the range of competition in party systems, whether executives are parliamentary or presidential, whether state power is centralized or dispersed, and so on. These institutional structures have significant consequences for patterns of political participation such as levels of voting turnout and types of election campaigning, as well as in rates of party membership and activism.[14] If digital technologies adapt chameleon-like to existing political systems, then we would expect to find considerable cross-national differences around the globe. The rapid adoption of the Internet as a lobbying and fund-raising tool in American election campaigns, for example, may reflect the particular form of interest group pluralism and money-driven political campaigns characteristically found in the United States, rather than a model common in many European democracies. The German SDP intranet, as a democratic mass-branch organization, may provide far

more opportunities for horizontal interaction and communication among party members than is available in the Japanese leadership-dominated LDP. The online delivery of services for housing and health in Swiss local cantons, with a stronger tradition of decentralized governance, may prove more advanced than equivalent services provided by English county councils. The parliamentary website for the Norwegian Storting can be expected to be far richer and more interactive than those designed for less influential and democratic bodies such as the Jordanian National Assembly or Thai Ratha Sapha. And so on. Despite the Internet's growing importance, at present little systematic empirical research compares its spread across nations and its functions in different political systems across the globe. This raises a series of issues: Who surfs in Germany, Japan, and Mexico? Who reads online newspapers or uses broadband television and radio in the United States, Taiwan, and Italy? What information about government services is available in Switzerland, Canada, and South Africa? How is the World Wide Web utilized by parties, by networks of alternative social movements, or by lobbyists in France, Sweden, and India? How is email employed to mobilize dissident groups, human rights activists, and opposition movements to challenge the authority of authoritarian regimes in Burma, Afghanistan, and China? Scattered case studies of digital politics are available in many particular countries, and a burgeoning literature is available in the United States, but so far the broader picture across the globe remains unclear.

TRIANGULATED RESEARCH DESIGNS

Another major challenge is that research on the Internet needs to integrate research findings drawn from numerous disciplines including those of communications, sociology, anthropology, history, social psychology, market research and business studies, computer studies, and industrial design, as well as political science.[15] Qualitative methodologies deconstructing the meaning of digital communications include discourse analysis, literary criticism, rhetorical studies, and textual analysis. Quantitative approaches include the standard techniques of sample surveys representative of the general population and special surveys of the online community, content analysis, focus groups, experimental research designs, and newer market research procedures monitoring user behavior like "click stream data" from cookies measuring activity on websites.[16] Yet in the early years, the available data meas-

uring Internet use often remain "guesstimates," even with the latest available market research techniques.[17] No single methodology can hope to capture the rich complexities of life on the Internet and this study therefore draws on hundreds of studies from different disciplines, as well as empirical evidence from aggregate data, content analysis, and cross-national surveys. The most effective research strategy is to triangulate among diverse sources of evidence, attempting to understand the Internet by piecing together a range of independent studies to see if the evidence points in a consistent direction across different countries. Where the findings conflict, we need to point out the uncertainties and consider some of the reasons leading to these different results. Where the results survive replication, this increases confidence in the reliability of the generalizations.

The book draws upon multiple databases to compare the worldwide patterns of use. Estimates of the online population are provided by NUA, a company that monitors surveys from a wide range of different market research companies.[18] Most data are collected for commercial purposes, to gauge the market for e-commerce. Although use of different surveys limits the reliability of the comparison, nevertheless this source provides the most comprehensive and up-to-date picture of Internet penetration rates worldwide. As discussed in the next chapter, evidence from independent sources serves to confirm the global pattern established in the NUA data, including information about the geographic location of Internet hosts, collected by many international agencies, and data on the distribution of telephones and computer equipment. Worldwide data on websites established by parliaments, parties, government departments, the news media, and interest groups are assembled from multiple sources, providing a comprehensive global map of digital politics. The study compares 179 nation-states worldwide, including in total 5.77 billion people living in societies at all levels of human and political development.[19]

Representative surveys in the United States and the fifteen-member states of the European Union are used to compare the social background, political attitudes, and behavior of the online community with the general population in these countries. In America, the first occasional opinion poll items on use or awareness of the Internet occurred in mid-1994 but it was only the following year that the population started to be monitored more systematically. The benchmark survey data used in this study are drawn from 1995–2000, which allows us to examine the rapid diffusion process in the emergent era as use of the new information

technology penetrated the United States and Western Europe. Broader comparisons would have been desirable but unfortunately systematic and reliable cross-national surveys measuring the impact of digital politics are still unavailable in most countries, and this has to await further research. The book draws on the series of American surveys conducted since 1995 by the Pew Center for the People and the Press, and also the National Election Study since 1996. For a broader comparison with the fifteen member states of the European Union, the book analyzes the biannual series of Eurobarometer surveys since 1995. Although identical items are not always available, functionally equivalent items allow comparisons to be drawn between America and Europe.

To understand digital politics within each country, the study analyzed the contents of a selected range of parliamentary and party websites around the globe. The aim was not to develop a comprehensive mapping exercise but rather a more limited attempt to isolate and compare some of the key functions of these sites. Using standardized instruments, the coding monitored the presence and depth of *informational* features (both text and graphics), and *communication* functions (such as opportunities to email the organization and its representatives, link to listservs, bulletin boards, and chat rooms, and other ways to become active). To analyze government websites, comparable data were drawn from the CyPRG group database, which has monitored the content and functions of official departmental sites worldwide since 1997.

Finally, as discussed earlier, the analytical framework used in this study (illustrated in Figure 1.1) distinguishes three nested hierarchical levels of analysis: the *macro-level* technological and economic environment which determines the availability and social distribution of Internet access within each country; the *meso-level* context of political institutions which provides the structure of opportunities mediating between citizens and the state including parties, parliaments, government departments, interest groups, new social movements and the news media; and *micro-level* individual resources and motivation affecting patterns of online civic engagement. This approach requires an analysis of both institutional and individual data. The framework assumes that levels of technological diffusion, such as the proportion of the population online, influences how political institutions have adapted to the Internet environment. In turn, the core institutions of representative democracy that are available in the digital world provide the systematic context within which citizens have opportunities to participate online. Which citizens choose to take advantage of these opportunities is

Not To Be Taken Out
Reserve Reading Room

understood to be determined by their resources (like time, money, and skills) and motivation (like interest, confidence, and efficacy). To develop this framework further, we can go on to examine which nations have emerged at the forefront of the knowledge economy, and which remain laggards in Internet diffusion, and the reasons behind these disparities at macro-level.

Not To Be
Reserve Reading

CHAPTER 3

Not To Be
Reserve Reading

Wired World

The World Bank, the United Nations, and the G-8 have expressed alarm that poorer societies lacking technological investment will drift farther behind their wired rivals in the global marketplace, whereas advanced industrialized societies will surge even farther ahead on the back of dramatic productivity gains. Multiple policy initiatives have been proposed, such as investment in technological infrastructure in Malaysia, computer training and education in schools in Latvia, and innovative community-level schemes in Bangladesh. Yet understanding the role of the state and the market in this process, and predicting which initiatives will succeed or fail in widening access in poorer societies, remains difficult unless we understand the reasons for the North–South divide. To unravel this issue we need to map the global spread of the information society and analyze the underlying conditions driving the process of technological transfer.[1] Many studies by historians, development theorists, and communication scholars have attempted to characterize the mechanics of the diffusion process, and economists and marketing specialists have attempted to identify the driving factors behind the demand for new products.[2] Drawing on this literature, after discussing the theoretical debate, this chapter focuses on four interrelated questions:

- What is the global pattern of Internet diffusion?
- Does this pattern represent the particular characteristics of Internet diffusion per se, or does it reflect similar trends found in the adoption of older forms of info-tech, such as radios, telephones, and televisions?
- In exploring the reasons for inequalities of Internet access, how far do cross-national differences reflect basic economic divisions

between rich and poor societies, so that we can predict the uptake of info-tech from standard economic indicators like level of per capita GDP and investment in R&D?
- Finally, what noneconomic factors determine technological diffusion across societies, including the role of human capital and democratic development?

In all cases we are interested in mapping and exploring the patterns common across different nations to establish the underlying conditions facilitating Internet access. But we also want to understand the exceptions to these generalizations, that is, those countries that have adapted to the information society far more successfully than would be predicted by their level of socioeconomic development alone. The outliers provide important clues to successful state interventions and market conditions.

THEORIES OF THE GLOBAL DIVIDE

International agencies have sounded the alarm over worldwide inequalities in the information revolution. UN Secretary General Kofi Anan warned of the danger of excluding the world's poor from the Internet: "People lack many things: jobs, shelter, food, health care and drinkable water. Today, being cut off from basic telecommunications services is a hardship almost as acute as these other deprivations, and may indeed reduce the chances of finding remedies to them." James D. Wolfensohn, president of the World Bank, has stressed the need to bridge the technological gap between rich and poor nations. "The digital divide is one of the greatest impediments to development," he argued, "and it is growing exponentially." The poorest societies face fundamental problems of basic survival and multiple difficulties with nutrition, literacy, and health, whether Mozambique ravaged by floods, Ethiopia decimated by ethnic conflict, or Zimbabwe plagued by AIDS. But for developing countries rising above the minimum economic threshold, such as Taiwan, Malaysia, and Brazil, access to information technologies has become important for integration into the global economy.[3] Cyber-optimists believe that info-tech has become a vital engine of growth for the world economy enabling many enterprising firms and communities to address economic and social challenges with greater efficiency. In poorer villages and isolated communities, a well-placed computer, like a communal well or an irrigation pump, may become another development tool, providing

essential information about storm warnings and crop prices for farmers, or medical services and legal land records for villagers.[4]

Cyber-optimists agree that the current situation shows a sharp North–South global divide but they believe that these inequalities will gradually fade over time, although not wholly disappear, under certain conditions: if access costs decline through falling prices for microprocessors and components and the growth of cheaper hand-held digital devices, mobile phones with Web-enabled technology, and community centers/Internet cafés; if the contents of the Web gradually diversify to become everybody's local radio and newspaper, community telephone exchange, and world marketplace; and if innovative programs by governments, nonprofits, and the private sector succeed in widening access to info-tech in developing nations.[5] A recent OECD report on the outlook for information technology concluded that industrialized countries account for more than 80 percent of the world market for information and communication technology, nevertheless expenditure in non-OECD countries has been growing at more than double the OECD average, with especially rapid expansion in telecommunications and IT hardware investments in Brazil and China.[6] Carlos Braga presents an optimistic scenario for developing countries' participation in the emerging knowledge economy: "Although, no doubt, income and wealth inequality may increase in the initial stages of the process, catch-up can also happen at a much faster pace than in the past. ICT spending, for example, grew more quickly in most developing regions than in high-income economies in the 1992–97 period. And countries like South Africa and Brazil already boast a higher share of networked personal computers than most industrialized economies."[7]

Numerous examples can be cited to show the expansion of digital opportunities in developing societies around the world.[8] Many Southeast Asian nations seek to emulate the Japanese model in the postwar era of reconstruction, and the knowledge-based economy in Singapore, South Korea, and Taiwan. In Malaysia, for example, the Multimedia Super Corridor has been developed to bring investment from telecommunications, multimedia, and electronics companies, and the production of silicon wafers and software. The corridor has attracted major players such as Microsoft, Sun Systems, and NTT (Japanese telecom). Under the "Vision 2020" plan, Malaysia now boasts cellular telephone penetration rates of one in every ten people, more and more wired schools, and 21 Internet hosts per 1,000 people. Revenue generated by the production of information and communica-

tion technology goods, such as office equipment, telecommunications, and consumer audiovisuals, shows that the United States leads the world but many Asian countries are close rivals, including Japan (second), Korea (third), Singapore (fourth), Taiwan (seventh), and Malaysia (eighth).[9] Southern India is most often cited as an important area of software development, producing an estimated $3.8 billion in revenues, with this figure doubling in the past few years. More than one-half of India's software services are exported to the United States.[10] The Bangalore area has attracted inward investment from many major corporations, not least from the diaspora of the Asian dot.com entrepreneurs thriving in California's Silicon Valley and Cambridge's Technology Park.[11]

In rural Bangladesh, many isolated communities lack landline telephones. An innovative program by Grameen Telecom supplies cellular mobile phones to village women, who rent calls in their community to repay the loan and sustain thriving microenterprises.[12] With this service, local communities benefit by direct links to job, weather, and health information, as well as more efficient markets for their produce. Village Telecom Centers are being developed with email and fax services, along with computer literacy projects in selected school.

In Europe, Slovenia, Estonia, and Slovakia have made great strides in getting their populations online, moving well ahead of Portugal, Greece, and Austria in levels of connectivity. Hungary's ambitious Schoolnet program has allowed students in two-thirds of all secondary schools to browse the Web from their classrooms, with extensive teaching resources, interactive discussion forums, events, and competitions.[13] The Estonian government has provided public access points for the Internet throughout the country, using schools, post offices, community centers, libraries, police stations, and health clinics. The program has been highly successful; today more than one in ten Estonians is online, with personal computer ownership well above average for Central and Eastern Europe.[14]

Progress has been slower in Africa, but nevertheless Global Crossing, Lucent Technologies, and Africa One plan for an ambitious $1.9 billion project to link up the entire continent by 2002 through a high-speed underwater fiber-optic cable, with interior countries connected through terrestrial cables, microwave, or satellite facilities, overcoming many of the current problems of the inadequate telephony infrastructure.[15] Given a high-speed backbone, and market liberalization of telecommunication services, African nations may also be able to

"leapfrog" stages of industrialization through new technology by investing in fully digitized telecommunications networks rather than outdated analog-based systems. Cellular telephony is rapidly expanding as an alternative to conventional network services; the number of subscribers in the OECD region reached almost one-fourth of the population in 1998.[16] This growth has had even greater impact in the developing world. In postindustrial economies there were 20 times as many mobile phones in 1998 as there were in 1990, and in developing economies there 160 times as many, an astonishing rise.[17] More than one-third of all telephone subscribers in Cote d'Ivoire, Cambodia, and Paraguay, for instance, are now connected via mobiles, a far higher proportion than in the United States.[18] But which of these schemes for promoting access to new technologies will succeed or fail? Do international agencies, nonprofit organizations, and national governments need to intervene or will the market eventually widen access? And what are the underlying social, economic, and political factors driving diffusion? To examine these questions we first need to establish which societies have moved ahead in the Information Society and which have lagged behind.

WHERE ARE PEOPLE ONLINE?

As yet no official government statistics on the online population are collected by international agencies such as UNESCO and the International Telecommunications Union (ITU), although indirect measures of technological diffusion are available, including investment in scientific research and development, the spread of computer hardware, and the rate of telephone density. The most comprehensive worldwide guide estimating the size of the online population is provided by NUA. This organization regularly monitors and collates survey results conducted by different market research companies in each nation. The surveys ask a representative sample of the public in each country about use of the Internet from home, work, or elsewhere during the previous three months. NUA's database *How Many Online* currently collects data from 179 countries, covering 5.7 billion people.[19]

A word of caution is needed about these estimates. First, the measure of Internet use is simple and limited. Surveys only monitor whether people were online, not how many days a week, still less how long people used the Internet every day.[20] Yet patterns of use can vary widely among those who occasionally click on the Web as a reference source in a public library or Internet café, and others constantly online via a 24/7

broadband cable link from home and a high-speed LAN connection at the office. Some include "regular users," others anyone who has accessed the Internet within the past year or month. Moreover, without a common methodology, the reliability of these estimates depends on the quality of the individual surveys. Market research companies vary substantially in their fieldwork, questionnaire design, and sampling methods – for example, in how far they include rural and illiterate populations. Different responses in surveys can be produced by even modest variations in question framing, wording, or order. Some data collected by NUA may therefore represent little more than approximate "guesstimates," especially in poorer countries where market research companies have little track record. These are important limitations, but fortunately there are two ways to check the consistency and reliability of the available data. First, NUA maintains a record of all survey results within the database. This means that we can compare the estimates provided by different companies in each country around the same dates. Where two or three data points were provided within a three-month period, the results were averaged to reduce random fluctuations by outliers. Even more important, the reliability of figures can be cross-checked against independent indicators by comparing the survey estimates of the online population with the distribution of Internet hosts and computers in each country. If a high correlation exists between independent sources this increases confidence in the reliability of the estimates.

Systematic trend analysis is available from 1995 onward. Although networked computing and computer-mediated email have existed for the scientific elite since the early 1960s, the number of users was too small to monitor through mass surveys. As discussed in the previous chapter, the key historic development transforming the Internet was a series of rapid innovations: the birth of the World Wide Web (1990) and the launch of popular browsers to access materials including those by Mosaic (1993), Netscape Navigator (1994), and Microsoft Internet Explorer (1995). Subsequent technological applications, such as the easy transfer of .mp3 music files and video formats, and WAP-enabled digital telephony, while representing important innovations, cannot yet claim to have had an impact equal to the basic invention of point-and-click browsers. Rather than a slow and steady process of evolution, development of the Internet has been one of modest incremental spread for thirty years, then a punctuated break in the early 1990s, followed by a rapid surge following the 'S' time curve among early adapter

nations. The NUA evidence highlights the dramatic rise in popularity of the Internet in recent years: between 1995 and 2000 the total number of Internet users surged from about 26 to over 400 million worldwide, an explosive jump within the space of a few years. The Internet became a truly global phenomenon as more and more users came online from around the world and the proportion of Americans in the online community dropped from 70 percent to 38 percent during 1995–2000 (see Figure 2.1). Despite this remarkable expansion, today about one in twenty of the world's population is online, with highly uneven diffusion worldwide.

The diffusion of the Internet today shows dramatic contrasts between and within regions of the world. Table 3.1 and Figure 3.1 display the proportion of the population online, estimated from NUA figures for November 2000 in 179 nations. Among the early adopters, Scandinavia and North America lead the world in rates of Internet penetration, followed by Western Europe, with about one in five online. Central and Eastern Europe, Asia, the Middle East, and South America fall below the world average, all with fewer than one in ten online, while minimal diffusion is evident in Sub-Saharan Africa. In terms of levels of human development, Table 3.1 highlights the stark contrasts between rich and poor nations. In 2000, most of the world's online community (84 percent) lives in highly developed nations.[21] In comparison, the thirty-five societies classified by the UNDP with low levels of human development, such as Nigeria, Bangladesh, and Uganda, contained about 1 percent of the online population, although home to half a billion people.

A finer-grained comparison of countries ranked in Figure 3.2, excluding nations where less than 2.0 percent of the population were online, reveals a pattern of widespread adoption in four clusters of societies: throughout the *smaller Nordic social democratic welfare states*, especially Sweden, Norway, Iceland, and Finland; in larger *Anglo-American and English-speaking nations* including the United States, Canada, Australia, and the United Kingdom; in the *Asian "tiger" economies* of Singapore, South Korea, and Taiwan, as well as Japan; and in a few *smaller European nations* with above-average Internet use such as the Netherlands, Belgium, Switzerland, Slovenia, and Estonia. The twenty-nine OECD member states representing industrialized nations dominate Internet users yet there are substantial contrasts even among neighboring countries within this category, for example between the United States and Mexico, or between

Table 3.1. *Internet users by region and type of society, 2000*

	Total population 1997, UNDP	Total online 2000	Total weighted hosts 2000	Number of nations
Industrialized	1,098,620,000	339,259,000	65,785,669	29
Nonindustrialized	4,672,773,000	66,237,700	2,161,528	150
North America	396,400,000	169,620,000	46,123,871	3
Asia-Pacific	3,270,250,000	99,463,200	5,950,661	35
West Europe	362,960,000	86,655,000	11,318,161	16
Central and Eastern Europe	404,180,000	16,310,500	833,419	13
Scandinavia	24,000,000	12,354,000	2,711,261	5
South America	403,893,000	13,710,000	748,279	30
Africa	685,020,000	2,956,500	21,608	52
Middle East	224,690,000	4,427,500	239,936	14
High development	1,011,650,000	340,810,500	65,517,356	43
Medium development	4,102,603,000	57,762,500	1,819,652	89
Low development	583,250,000	422,200	4,520	35
Total	**5,771,393,000**	**405,496,700**	**67,947,197**	**179**

Notes and Sources: **Total Population** 1997 from the United Nations Development Report, 1999, New York: UNDP/Oxford University Press; **Number online** in 2000 or latest date estimated from *www.NUA.ie*, "How Many Online?" and from the International Telecommunications Union Basic Indicators 1998, *www.itu.int*; **Number of weighted hosts** estimated by the International Software Consortium January 2000. The sites for .com, .net, and .org were weighted according to the location of registration and reallocated by nation according to the OECD methodology. See OECD 1999, *Communications Outlook 1999*, Paris: OECD, *www.oecd.org*. **Level of human development** defined by the Human Development Index 1999, *Human Development Report, 1999*, New York: UNDP/Oxford University Press.

Australia and New Zealand, as well as between Italy and Greece. European Union member states are all affluent, advanced economies with a large service sector, a well-trained workforce, and high levels of personal income. Nevertheless following rapid growth in Internet usage in the late 1990s, there is widespread diffusion today in Sweden, Denmark, and Finland, moderate use of the Internet in the UK, Germany, and the Netherlands, but few online in Greece and Portugal,

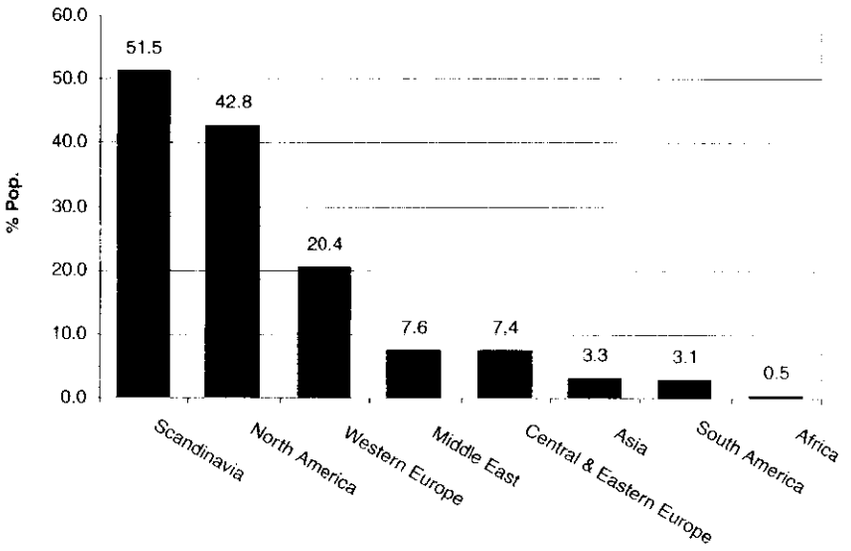


Figure 3.1. The Percentage of the Population Online by Major Region, 2000.
Source: "How Many Online?", *www.NUA.ie*, November 2000.

the latter ranking well below developing societies such as Slovenia, Estonia, and Taiwan.[22] At the bottom of the national rankings, with less than 2.0 percent of the population online, few Internet users are found throughout most of the poorer countries of Sub-Saharan Africa (with the exception of South Africa), as well as in many states in Central Europe, the Middle East, Asia, and Latin America.

The predominance of the United States on the Internet has gradually declined over time as more and more people have come online elsewhere, although still one-third of all users worldwide are American (154 out of 406 million users). The vast region of Asia-Pacific, including the massive populations of China, India, and Indonesia, contains another 99 million users, and 86 million live in Western Europe. In contrast Sub-Saharan Africa, home to more than half a billion people, contains only 3 million Internet users, or less than 1 percent of the world's online community.

To check reliability, these estimates can be compared with the geographic location of Internet hosts based on data provided by agencies such as Network Wizards, the Internet Software Consortium, and Netcraft. To function as part of the Internet all computer hosts require a domain name (like *www.yahoo.com*) and an associated Internet Protocol address record (in numeric form such as 193.51.65.17). Domain Name

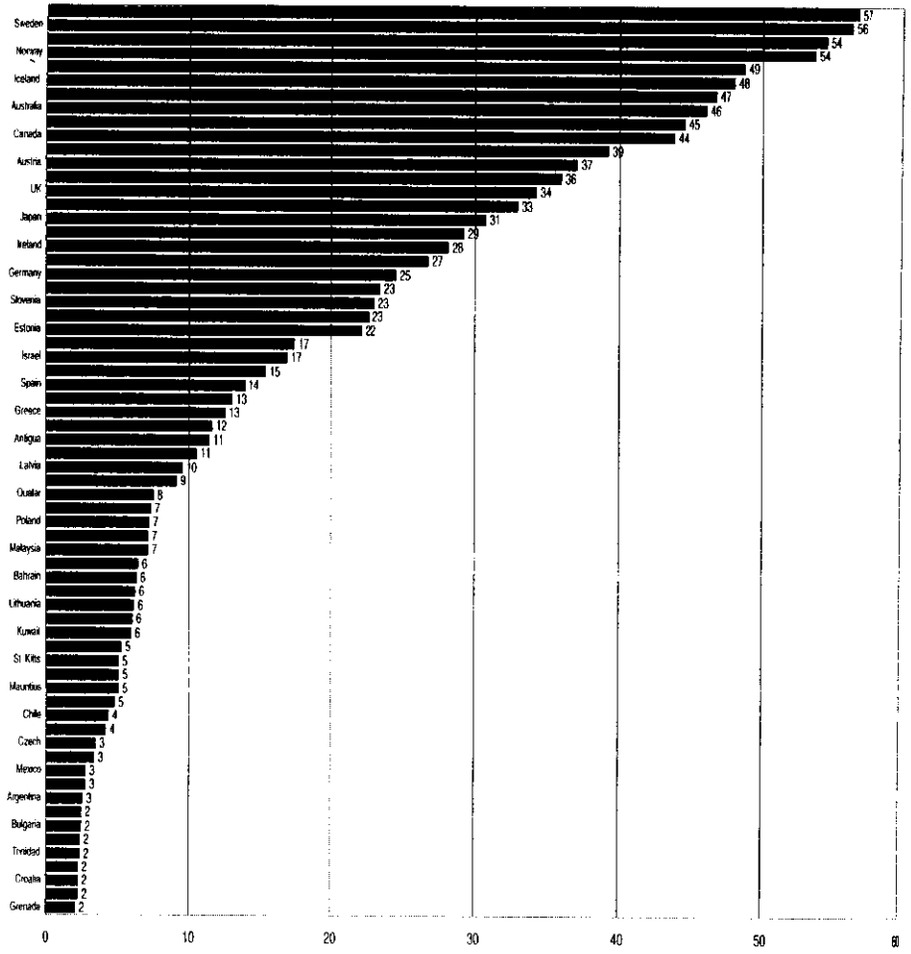


Figure 3.2. The Percentage of the Population Online by Nation, 2000.
Source: All countries with more than 0.5 percent of the population online. "How Many Online?", *www.nua.ie*, November 2000.

System (DNS) servers translate back and forth automatically between names and numbers to locate a site. Domain names are divided into two categories: *national* top-level domains such as those ending in .fr (France), .uk (United Kingdom), or .be (Belgium), and *generic* top-level domains like the dot.coms, dot.nets, and dot.orgs. The analysis of domain names provides the best available estimates of the location of Internet hosts although there are some important limitations to these data as well.

The existence of "company firewalls" means that the figures provide a minimum estimate of the number of hosts and it remains impossible to tell how many users are online via each host. Another major problem with the generic domains (like dot.coms) is that it is difficult to analyze them on a geographic basis. Following the methodology used by the OECD, this difficulty can be overcome by allocating them on a weighted basis according to the country of registration of generic domains.[23]

The maps in Figure 3.3 confirm the pattern of global inequalities already observed. The distribution of hosts according to the Netcraft data is strongly correlated with the proportion of the population online in each country provided in the NUA estimates ($R = .854$ Sig. .001), increasing confidence in the reliability of the data sources. The comparison again shows the striking predominance of a few countries including Finland, the United States, Norway, Iceland, Canada, Sweden, and Australia, and the significant absence of hosts and online users throughout large swathes of the rest of the world. There are almost as many hosts located in France as in all of Latin America and the Caribbean, and there are more hosts in New York alone than throughout Sub-Saharan Africa.[24]

RELATIVE INEQUALITIES IN THE INFORMATION SOCIETY

The fact that there are *absolute* inequalities between rich and poor nations in the virtual world is hardly surprising; it would be naive to expect otherwise given the substantial disparities in every other dimension of life from health care and nutrition to education and longevity. Despite the more exaggerated hopes of some cyber-optimists, the Internet is not going to suddenly eradicate the fundamental and intractable problems of disease, debt, and disadvantage facing developing countries. The more interesting question, with important implications for understanding the new media, concerns the *relative* inequality of opportunities. Is it easier or more difficult to go online in different societies, compared with inequalities of access to other types of communication technologies, such as telephones and televisions?

On the one hand, the pattern we have established in the distribution of the online population may represent a particular characteristic of Internet diffusion per se. If this is the case, then it would be appropriate to search for possible causes related to this particular form of technology, such as the costs of computer hardware, software, and ISP connection charges, the need for computing and literacy skills to go online, and the predominance of English-language contents. The availability of

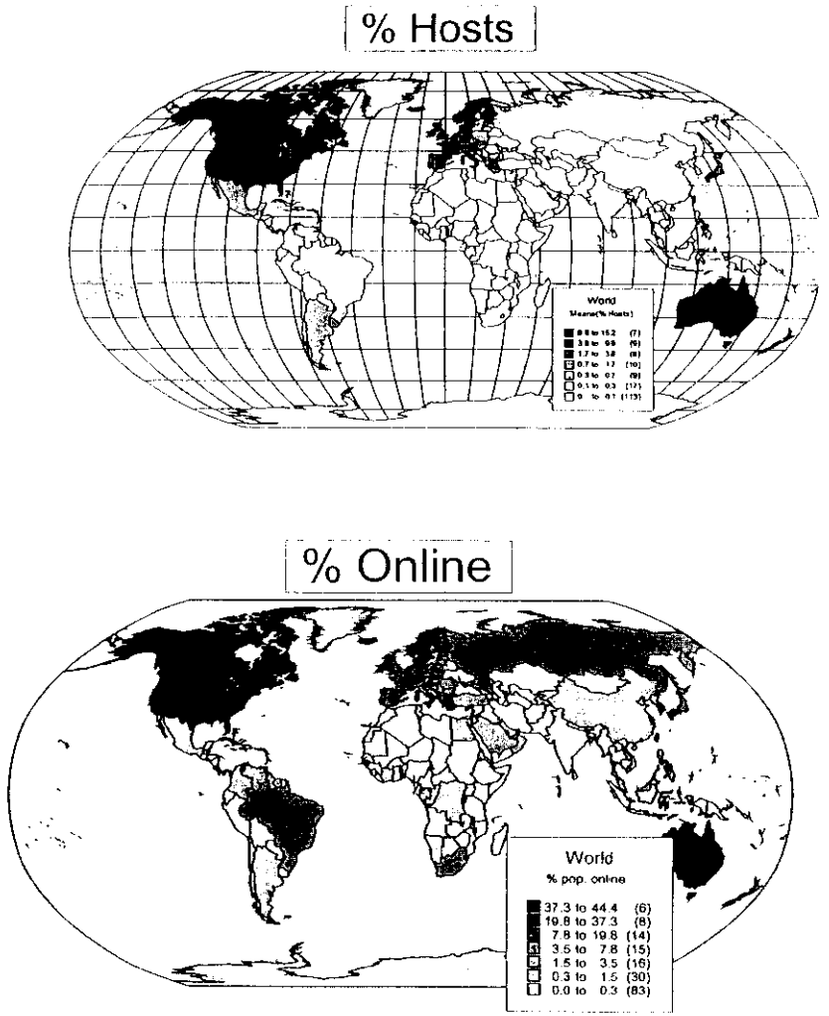


Figure 3.3. The World of Internet Hosts and the Online Population, 2000. *Sources:* **Online Population:** "How Many Online?", *www.NUA.ie*; **Internet Hosts:** *www.Netcraft.com.*

computers is often assumed to be one of the most fundamental barriers to current use of the Internet for most people. Worldwide the number of personal computers tripled in the last decade, from 25 PCs per 1,000 people in 1990 to 71 in 1998. But again the regional disparities are marked: There are 459 computers per 1,000 people in the United States but only 7 per thousand in Sub-Saharan Africa.[25] Most Internet connections in developing countries still travel down mainline telephone lines so the rate of teledensity would also be expected to be related to patterns of Internet access. If this analysis is correct, it suggests that the solution lies in equalizing technological resources, such as through the distribution of computer hardware, skills training, and network connections.

On the other hand, the global patterns of inequality in Internet use may mirror the broader pattern of access to the Information Society. If so, we would expect to find a strong correlation between the spread of digital technologies and the availability of traditional mass media that have been around for decades or longer, such as newspapers, telephones, and television. If this is the case, it suggests that the problems of Internet access in the developing world are not particular to the nature of the medium itself – you don't need keyboard or literacy skills to switch on a radio – but instead may be due to deep-rooted and endemic problems in poorer societies such as the general lack of income, leisure time, literacy, and education that hinder use of traditional media such as newspapers. If this is the case, rather than any short-term fix, such as delivering beige desktop PCs to wired schools in Mozambique, Egypt, and Bangladesh, the long-term solution would be general aid, debt relief, and economic investment in developing countries. Understanding these issues can provide important clues to the fundamental reasons underlying the process of technological diffusion and therefore the most effective policy initiatives to overcome digital inequalities.

To analyze these questions, the different types of information and communication technologies can be divided into two categories. A standardized 100-point *New Media Index* was calculated by combining the proportion of those online within each country, as already described, with the per capita distribution of hosts and of personal computers. We would expect a strong correlation between these factors. A similarly standardized 100-point *Old Media Index* combines comparable per capita data in each country measuring the distribution of daily newspapers, radios, television sets, mainline telephones, and mobile phones (see Table 3.2). For an overall summary, multiplying both measures and dividing by two created the composite 100-point *Information Society Index.*

The results show two important findings. First, use of all forms of communication media is highly intercorrelated, meaning that countries at the forefront of the Information Society on one indicator are likely to lead on many others as well (see Table 3.3 and Figure 3.4). Info-rich countries like Sweden, the United States, and Australia are not just ahead in

Table 3.2. *Proportion of the population using new and old media*

	New Media			Old media					
	Population online	Weighted hosts	PCs	Radios	TV sets	Daily newspaper	Mainline phones	Mobile phones	Info-Soc index
	2000	2000	1998	1997	1998	1996	1998	1998	
Region									
Scandinavia	35	11	36	112	58	45	64	47	80
North America	27	10	28	118	61	16	47	15	63
West Europe	12	3	27	79	53	21	53	24	52
Central and Eastern Europe	3	0.3	6	45	32	13	21	4	30
Asia-Pacific	5	1	8	35	19	11	13	8	27
Middle East	3	0.2	6	39	25	11	19	8	22
South America	1	0.1	5	38	22	8	15	3	18
Africa	0.3	0.1	1	17	5	1	3	0.5	6
Development									
High	14	4	23	83	49	26	46	23	53
Medium	1	0.1	3	32	21	6	11	2	15
Low	0.02	0.03	0.3	14	3	1	1	.01	4
Total	**4**	**1**	**9**	**40**	**24**	**10**	**18**	**7**	**38**
No. of Nations	169	179	125	140	139	133	179	139	101

Notes: All figures are expressed as a percentage of the population. For full details see Table 3.1. Percentage of radios (1997), television sets (1998), mobile phones (1998), and PCs (1998) from *World Development Indicators 2000*, The World Bank. Level of development is classified according to the UNDP (1999). The 100-point Info-Soc Index is calculated by combining all indicators into a standardized scale.

Table 3.3. *Correlations in use of the new and old media*

	New media				Old media		
	Online	Hosts	PCs	Radio	TVs	News-papers	Mobile Phones
Hosts	.854						
PCs	.806	.745					
Radio	.788	.708	.818				
TVs	.692	.614	.769	.848			
Newspapers	.725	.715	.788	.749	.734		
Mainline phones	.791	.710	.886	.837	.861	.839	
Mobile phones	.809	.827	.845	.754	.715	.830	.872
InfoSoc Index	.883	.810	.924	.937	.917	.858	.954 .888

Note: See Tables 3.1 and 3.2 for details of the measures. The figures represent correlations (r). All are significant at the .01 level.

Sources: See Tables 3.1 and 2.2.

terms of the Internet but also in the distribution of other media such as newspaper readership, radio and television sets, personal computers, and mainline and mobile cell telephones. Correlation revealed that access to all these media fell into a single consistent dimension.[26] There was little distinction between use of old and new media; the proportion of those online in each country was most strongly related to the distribution of hosts, telephones, and PCs, but it was also significantly and strongly related to the distribution of radios, TV sets, and newspaper readership in each nation. This means that people living in poorer societies excluded from the world's flow of communications such as Burkina Faso, Yemen, and Vietnam were largely cut off from all forms of info-tech, including traditional mass media like radios and newspapers as well as modern ones such as mobile phones and personal computers. The vast majority of poorer societies, low on both indicators, cluster together in the bottom left corner of the scatter plot illustrated in Figure 3.4. At a broader level, as cyber-pessimists suggest, this analysis indicates that in the emerging Internet era, the relative inequalities between rich and poor in access to the virtual world reflect, rather than transform, global disparities in the

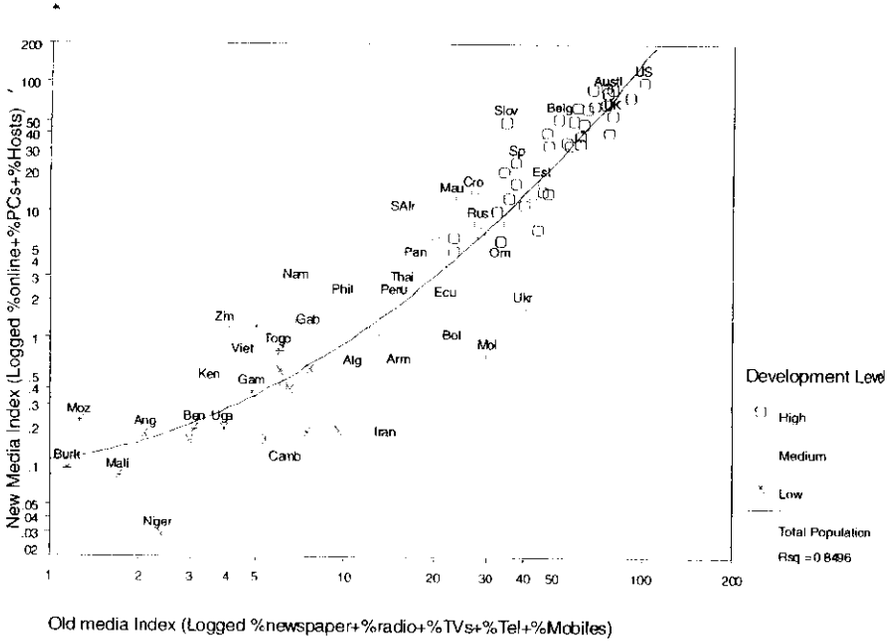


Figure 3.4. The Information Society.
Source: For details about the construction of the scales and the data sources see Table 3.2.

Information Society. Nations that already have many radios and TVs are most likely to also get access to networked computers.

The previous comparison also reveals important variations to this general pattern. Some new democracies such as Slovenia and South Africa fall above the regression line in Figure 3.4, indicating that these are more advanced in access to the digital technologies than would be expected from their use of television and newspapers. Some countries like Finland, Germany, and the United Kingdom fall exactly where predicted along the line, suggesting a balance in these societies between new and old media. The United States also falls into this category, leading the world in terms of connections to info-tech. Many less affluent societies like Bolivia, Iran, and the Ukraine, as well as Portugal and Greece, while lagging on both scales, are slightly stronger in use of the traditional print and electronic media than in access to digital technologies. Therefore although the analysis reveals that access to old and new media technologies is closely related, the outliers suggest some exceptions to the rule that may have significant implications for policy initiatives designed to broaden the spread of the wired world.

ECONOMIC DEVELOPMENT AND THE DIGITAL DIVIDE

How do we explain the diffusion of the Information Society that we observe? Many possible answers can be suggested. If the pattern reflects basic differences in levels of economic development, as many assume, then we would expect to find clear divisions between rich and poor countries. One of the most comprehensive studies seeking to explain the distribution of Internet hosts within postindustrial societies, by Hargittai, concluded that the economic wealth of a country, measured by per capita GNP, was one of the most important predictors.[27] A study for the International Telecommunications Union also found that the number of Internet hosts per country was significantly related to general levels of socioeconomic development, using the UN Human Development Index measuring the rate of adult literacy, education, life expectancy, and per capita GDP.[28] Rodriguez and Wilson's research for the World Bank arrived at similar conclusions.[29] If this pattern is confirmed in this study it would suggest that the Internet represents one more disparity reflecting the poverty of those living in developing nations, lacking access to the knowledge economy as well as basic nutrition, education, and health care.

The results of the regression analysis confirm that the relationship between Internet use and economic development, measured by per capita GDP in 1997, is indeed both strong and significant ($R = 0.77$ Sig. .000). The best-fitting model with logged data, illustrated with the scatter plot in Figure 3.5, shows that there is also a critical threshold: the online population expands exponentially once countries rise about the $9,000 level of per capita GNP. Not surprisingly, many of the poorest nations such as the Sudan, Rwanda, and Bangladesh are among the least-wired societies, whereas at the other extreme many affluent postindustrial economies such as Norway, Finland, and New Zealand fall exactly where expected on the regression line. Yet at the same time there are also some important outliers to the general pattern, and particular explanations about these countries may provide important clues to what other factors may contribute to Internet connectivity beyond economic development.

In one category, many countries, falling below the regression line, are relatively affluent yet display lower than average Internet penetration rates. This includes certain oil-rich Middle Eastern nations such as Bahrain, Kuwait, the United Arab Emirates, and Saudi Arabia, as well as South East Asian states such as Singapore, Japan, and Brunei. The reasons for this pattern cannot be established systematically here but they

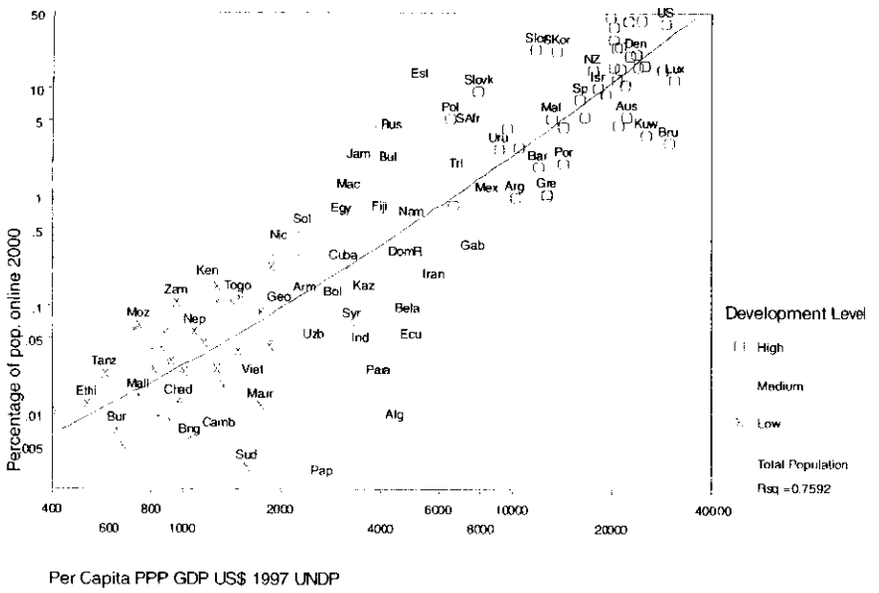


Figure 3.5. Economic Development and Internet Use.
Sources: **Per Capita GDP** in Purchasing Power Parity, United States Dollars, 1999: UNDP, *Human Development Report, 1999*, New York: UNDP/Oxford University Press; **Percentage of Population Online**: "How Many Online", *www.NUA.ie.*

can perhaps best be explained by the extreme inequalities of personal income and wealth characteristic of these societies, possibly combined with some cultural restriction on use through restricting Internet Service Providers in some Middle Eastern countries.[30]

In contrast, some other societies are above the regression line, with higher than average Internet access than would be predicted by their level of economic development, including Slovakia, Slovenia, Poland, and Estonia, as well as in South Korea and Taiwan. These exceptions to the general pattern have important implications for understanding this phenomenon, explored later in the analysis, because they weaken claims of economic determinism and highlight the significance of other factors leading to technological diffusion. Economic development certainly predicts access to the Internet, as it does to other traditional forms of information and communication, but the wider scatter of countries in the top right corner of Figure 3.5 shows that, once above a certain minimal level, economic development is not necessarily essential to greater online use.

Beyond economic development, other economic factors that might influence use of the Internet include the availability and costs of technology in different countries, including the price of computer hardware/software, connection charges through Internet Service Providers, and the provision and availability of mainline and mobile telephone services. This is not just a matter of the familiar income disparities between rich and poor nations, because the relative costs of ISP access are actually higher in many developing countries. Although some free ISP and email services are becoming available, the OECD estimates that average monthly charges for Internet access in July 1999, based on a 20-hour per month peak rate, were $24 per month in Australia but almost three times as much in Turkey ($65 per month), and almost four times as much in Mexico ($94 per month).[31] In OECD nations, the cost of ISP access was significantly correlated with the proportion online in each country ($R = -.45$ Sig. .01). Countries with relatively high Internet penetration rates (e.g., Finland and Sweden) often have lower access prices. ISP charges are gradually falling, making access more affordable for business and consumers, but the costs of local telephone connections may also be an important deterrent, especially in Europe compared with the United States. Hargittai's comparison of OECD member states concluded that telecommunications policy, notably the level of competition under deregulated systems, was significantly associated with levels of Internet connectivity.[32]

Another important reason for the spread of digital technology may lie in the broader process of research and development within each country, particularly basic investment in science and technology. One reason why use of mobile phones is so high in Scandinavia, for example, is because Nokkia and Ericsson, two leading telecommunications manufacturing companies, are located there. Expenditure on scientific and technological research and development can be gauged by many indicators, such as the proportion of scientists, engineers, and technicians employed in R&D, the proportion of science and engineering students in tertiary education, and the percentage of manufacturing exports from high-tech products. Most of these indicators are highly intercorrelated and information was not available from World Bank and UNESCO data for all the countries under comparison. In this study, therefore, the impact of the technological environment can be compared by examining the impact of per capita spending on research and development as a proportion of GNP within each country. The results are illustrated graphically with the scatter plot in Figure 3.6, which shows a consistent and significant

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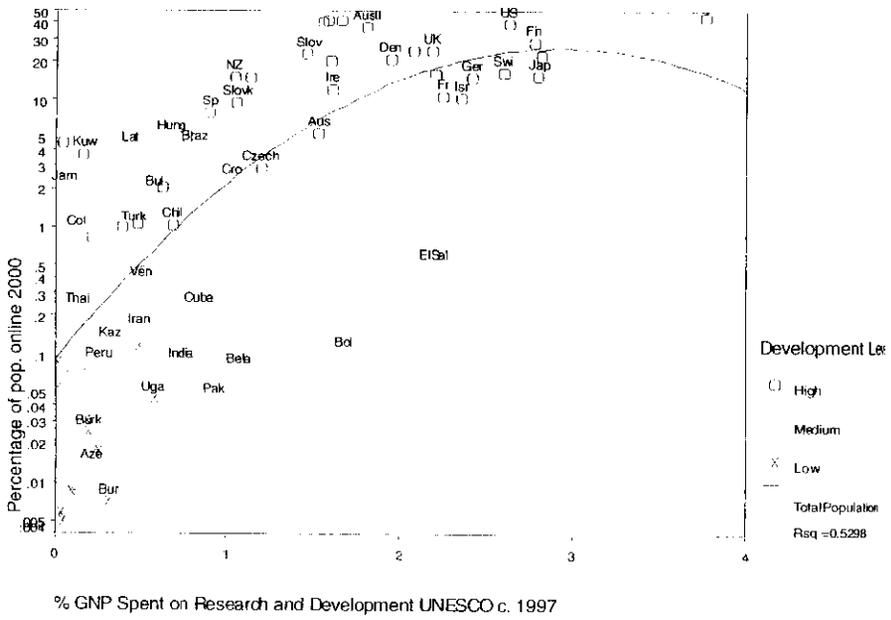


Figure 3.6. Research and Development Spending and Internet Use.
Sources: Percentage of GDP Spent on **Research and Development, 1997**: UNDP *Human Development Report, 1999*, New York: UNDP/Oxford University Press; **Percentage of the Population Online:** "How Many Online", *www.NUA.ie.*

correlation, confirming that R&D investments help boost the market for digital technologies, although there are a few outliers such as El Salvador and Bolivia, with lower than average online access given their level of spending on research and development.

HUMAN CAPITAL AND DEMOCRATIC DEVELOPMENT

Economic development goes a long way toward explaining the global disparities between richer and poorer societies; however by itself this fails to account for the substantial differences in Internet access between relatively similar countries, such as between the Nordic and Mediterranean European nations. What other answers could help to explain the patterns we have found? The evidence here remains more scattered but we can examine some common assumptions, especially the role of human capital and democratic freedom of expression.

The development of human capital – meaning the investment in digital skills and capacities through education, training, and lifetime learning – represents one of the most important factors that might facilitate Internet access. Human capital represents the properties that allow people to live and work productively. Education is one of the most significant forms of social development, producing the skills and experiences that are most likely to contribute to using computers. Secondary and university education provide direct hands-on training in basic and advanced computing techniques, familiarity with standard software packages, and confidence in surfing the Web, as well as the general cognitive skills necessary to make sense of the online information. Academic institutions may also play an important role in spreading digital technologies because they are often among the first institutions in a nation to be wired. Studies have found that more educated people are quicker to adapt to new innovations.[33] Education also contributes directly to basic literacy, and reading and writing skills that are currently essential to using this media. The World Wide Web is rich in visual imagery, such as pictures and streaming video, and audio content is burgeoning through the easy availability of online radio and .mp3 music. These types of content are likely to surge in popularity as broadband fat-pipe delivery increases in capacity, fiber-optic cables become capable of carrying more bandwidth, and file compression continues to shrink the storage size of data. Nevertheless most material now on the Web remains closer to text-based newspapers rather than to audio or visual electronic media like television. If basic illiteracy is an important barrier to access, most of the world's poor could still be excluded from the information revolution, even if computing equipment became more widely available through Internet cafés, schools, public libraries, and community centers. One-fifth of the world's population remains illiterate, and the total rises to 40 percent or more of those living in Sub-Saharan Africa, the Middle East, and Southern Asia, with even higher figures for the female population.[34]

Social development also encompasses language skills. Wresch suggests that knowledge of English as a first or secondary language is critical: "English dominates the world of computing. Both the English and the Americans like to claim their nation as the birthplace of computing, but in one way it doesn't matter which did what – both are English speaking nations."[35] A comprehensive analysis of more than one billion unique Web pages by the Inktomi search engine found that 87 percent of all documents were in English, a language understood by an estimated one in ten people worldwide.[36] This picture is confirmed by a more limited search in 1997 by the Babel team for the Internet Society which esti-

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mated that 84 percent of all pages in English, followed by German (4.5 percent), Japanese (3.1 percent), and French (1.8 percent), Spanish (1.2 percent), Swedish (1.1 percent), and Italian (1 percent), with all other languages each below 1 percent.[37] The technology to overcome language barriers is progressing, with online translation services freely available for the major European languages such as German, Spanish, and French, but nevertheless it is striking that many of the societies at the leading edge of the digital revolution are either Anglo-American countries like Canada and Australia, or smaller well-educated welfare states like Norway, Sweden, and the Netherlands where English has become the lingua franca as the second language taught widely in schools. By most standard economic indicators, major G-8 rivals such as France, Germany, and Japan should outrank the United Kingdom in terms of Internet penetration yet they continue to lag behind; for example, there are twice as many users in Britain than France, although this may also be influenced by the checkered history of the French Minitel project. Familiarity with English as a second language among the scientific elite may also help to explain the success of software development and computer manufacturing in ex-British colonies such as India, Malaysia, and Hong Kong. Rao et al. argue that elsewhere in Southern Asia the lack of Web content in indigenous languages, as well as the limited local relevance of the available sites, limits the incentive to go online for much of the population.[38] The predominance of the English language may change in coming years, as more and more of the population moves online in China, India, and Brazil, but in the meantime those who can read English are greatly advantaged in the digital world and, although we cannot prove the proposition here with the available evidence, this could plausibly help to explain high Internet penetration rates in the United States, Canada, Australia, and the United Kingdom.[39]

Finally, broader issues of *democratic development* may also contribute to digital diffusion through the expansion of opportunities for freedom of expression and civil liberties commonly available in democratic states. Freedom House has documented many instances where authoritarian regimes have attempted to limit public access to the Internet or censor its contents and freedom of publications on the Web. The study by Leonard Sussman found that although control of the Internet was far more difficult than censorship of television or the press, nevertheless currently about twenty states restrict citizen's access to the Internet to a significant degree.[40] Cited examples include "cyber-dissidents" jailed in China, the need for Burmese owners to report computers to the government, under penalty of a fine, and restricted ISP access to information content "contrary to Islamic values" in Saudi Arabia, Syria, and the United Arab Emirates It is difficult to monitor such incidence on a systematic basis but as an indirect indicator the impact of a country's level of democratization can be compared, using the Freedom House 7-point index measuring political rights and civil liberties, which has become adopted as the standard indicator for cross-national comparisons.[41] If democratization plays a significant role, then we would expect that countries with greater civil liberties and freedom of expression would display more widespread access to digital technologies.

The relative importance of economic development, human capital, and levels of democratization on rates of Internet penetration are analyzed in Table 3.4 using simple correlations and multivariate regression models.[42] Regional dummy variables were also included to see how far these remained significant after controlling for prior levels of economic development, human capital, and the democratic background of different areas. For comparison, alternative models (not reported here) were tested before arriving at the final one based on goodness of fit, parsimony, and the prior theoretical assumptions that determined the causal ordering of the variables.[43] The models were run to determine whether similar factors that predicted Internet penetration rates also helped to predict access to new and old media technologies.

Two important findings emerged from this analysis. First, the results of the simple correlations, without any prior controls, show that all the selected indicators of economic development, human capital, and democratization proved to be strongly and significantly related to use of the Internet, as well as to use of the new media and old media, in the expected direction. In other words, the proportion of the population online in each country was significantly related to levels of per capita GDP, R&D spending, literacy and secondary education, and level of democratization, according to the available measures. The regional patterns confirmed the significance of the differences observed earlier, even after controls, with higher than average use evident in Scandinavia, North America, and Western Europe, and significantly lower than average use in South America and Africa. Similar strong correlation coefficients were evident using the composite New Media and Old Media Indexes, suggesting a robust relationship that held up to testing across all indicators. The strength of the coefficients varied slightly; for example, patterns of literacy were more closely associated with use of the old media than the new, but the overall pattern was highly consistent.

Table 3.4. *Explaining the spread of technology*

	Percent online		New media index	Old media index
	Correlation	Beta (Sig.)	Beta (Sig.)	Beta (Sig.)
Economic development				
Per capita GDP	.74 **	.37 **	.57 **	.58 **
R&D spending (%)	.46 **	.32 **	.28 **	.32 **
Social development				
Adult literacy (%)	.40 **	.03	–.05	.09
Secondary education (%)	.53 **	.07	.01	.01
Political development				
Level of democratization	.55 **	.08	–.06	.10
Region				
Scandinavia	.59 **	.26 **	.19 **	.08
North America	.34 **	.17 **	.12 **	.04
Western Europe	.25 **	.07 *	.08	–.06
Middle East	–.04	–.07	–.11	.01
Central and Eastern Europe	–.05	–.06	–.03	.12
South America	–.29 **	–.11	–.09	.01
Africa	–.29 **	.01	–.03	.01
Adjusted R^2		.80	.57	.91

Notes: The dependent variables include the proportion of the population online (spring 2000), the New Media Index (% online, % PCs, and % hosts), and the Old Media Index (% newspapers, % radios, % TVs, % telephones, % mobile phones) (see Table 3.2 for details). The figures represent correlation coefficients (R) without any controls and standardized beta coefficients from multivariate OLS regression analysis models. The regional dummy variables drop Asia. *Sig. = –.01, **Sig. = .05.

Sources: See previous tables in this chapter for data sources.

Because each of these indicators is closely related to the other, these strong associations could prove to be spurious, for example if economic development simultaneously produces a more educated and literate public, higher levels of R&D investment, and the transition to a more democratic regime. There could well be a complex pattern of interaction here. To disentangle these factors we need to turn to the multivariate analysis models that assume that economic and social development come before the process of democratization. The most important finding from the regression models in Table 3.4 is that *economic factors outweighed all others in predicting cross-national differences in access to the information society.* Economic development, measured by per capita GDP, was consistently important across all three models, indicating that more affluent societies have access to a richer range of information and communication technologies, among both traditional and digital media. The poorest developing societies such as Mali and Nigeria lack old and new technologies of mass communications. Countries with per capita GDP in the middle region of about $6,000–10,000 show a more mixed picture; in particular, African societies tend to be poor in both indicators but Central and Eastern European nations have lower than average income but higher mass media use. Finally, affluent postindustrial societies including the United States, Norway, and Japan are high in both income and information resources. Moreover, the related economic indicator of levels of technological investment, as measured by per capita spending on research and development as a share of GNP, was also confirmed to be significant across all models, for reasons discussed earlier.

Once these related socioeconomic factors were included in the models, other indicators of human capital and democratization became insignificant. This suggests that literacy, education, and democratization do not exert an independent influence on Internet penetration rates, or access to older forms of information technology.[44] This pattern lends further confirmation to the findings of Hargattai's earlier comparison of OECD states.[45] After controlling for all these factors, the regions of Scandinavia, North America, and Western Europe still emerged as significantly ahead of all other areas in levels of digital connectivity, indicating that many other residual factors in these societies not specified in these models contribute to the diffusion process, such as policy initiatives to spread the Internet through wired schools, community centers, and businesses. The economic factors predicting Internet connectivity also serve to explain the diffusion of traditional mass media, suggesting a pervasive and robust pattern irrespective of the specific type of info-tech or the particular indicators selected for analysis. Of course the interrelationship of economic, social, and democratic development does create problems for these models, since it is difficult to isolate the influence of each. The models assume that economic development leads to human capital, such as a better educated workforce, but, of course, the reverse could be equally true. In the same way, although there is no one-to-one relationship, nevertheless consolidated democracies tend to be more affluent countries.

Another major limitation of the analysis is that the Internet is a relatively recent phenomenon, so it is possible that patterns of access evident today may change considerably within the medium- to long-term. One way to explore this further is to examine the global diffusion of traditional mass media such as radio and television, to see if the gap between rich and poor nations has gradually closed over time in these technologies. Based on UNESCO data, Figure 3.7 indicates trends in the adoption of television sets and radio receivers in developing and postindustrial societies during the last three decades. The results illustrate how these media have grown in popularity worldwide during the late 1980s and early 1990s. But, most important, the trends also show that the gap in access to these media between postindustrial and developing societies has not diminished in the last thirty years – if anything the reverse. Because use of television and radio grew more sharply in postindustrial societies in the 1970s, even with a slight plateau effect, the inequalities of access today between rich and poor nations are actually slightly greater than about three decades ago. If we can extrapolate from this pattern to digital technologies, this suggests that the relative disparities between developing and postindustrial nations that we have observed today will not necessarily close as more and more people go online. The prognostication is that early-adopter countries seem likely to maintain their relative lead, leading in digital technologies, even while laggard societies attempt to catch up.

CONCLUSIONS: NEW TECHNOLOGIES, OLD INEQUALITIES

International organizations including the World Bank and the UNDP have raised concern about the growing global digital divide in the 1990s, and the need to overcome this disparity before the situation rigidifies into a new virtual Berlin Wall splitting rich and poor worlds. Potentially digital technologies may serve the needs of the developing world. As well as the direct economic benefits for trade and tourism, the Internet may accelerate the longer-term investment in human capital. Information infrastructures for distance learning and educational development can facilitate access to the most up-to-date materials for teaching and research. As the World Bank Report expressed these hopes: "ICT greatly facilitates the acquisition and absorption of knowledge, offering developing countries unprecedented opportunities to enhance educational systems, improve policy formation and execution, and widen the range of opportunities for business and the poor. One of the

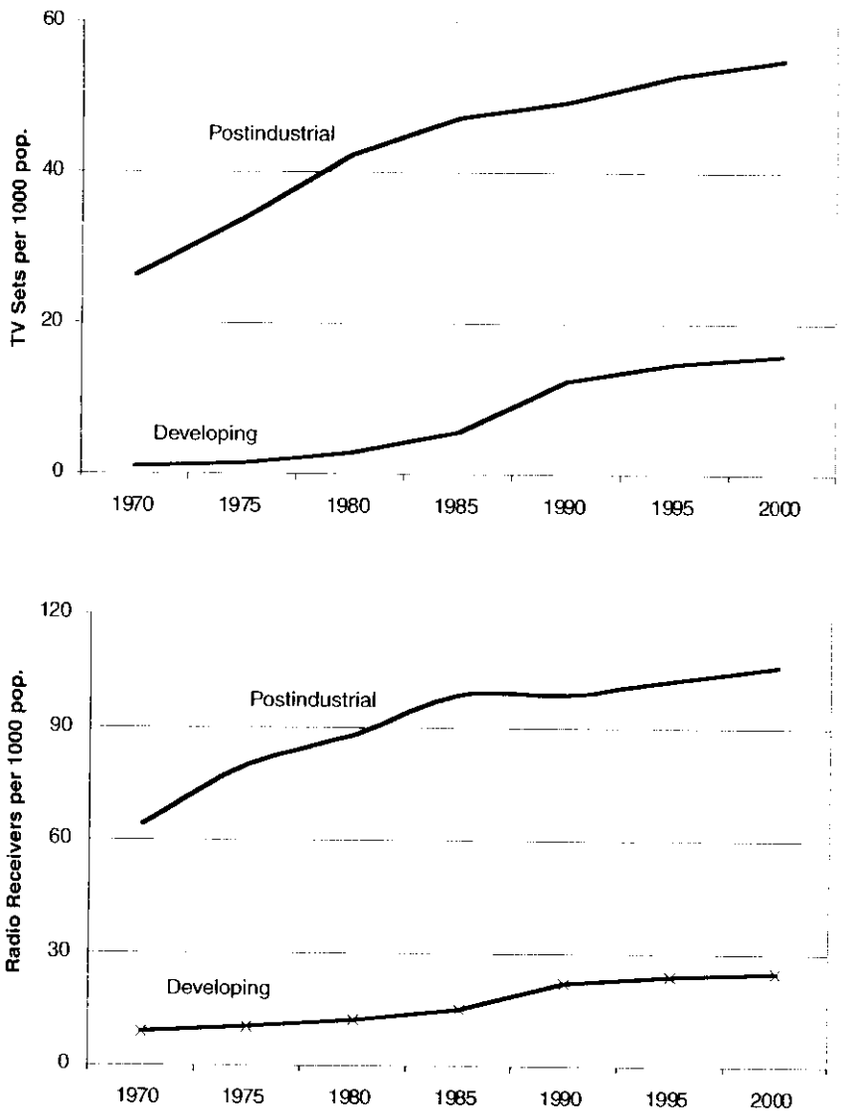


Figure 3.7. Trends in the Worldwide Spread of Televisions and Radios, 1970–2000.
Source: UNESCO Yearbooks, Paris: UNESCO.

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greatest hardships endured by the poor, and by many others who live in the poorest countries, is their sense of isolation. The new communication technologies promise to reduce that sense of isolation, and to open access to knowledge in ways unimaginable not long ago."[46] Hopes about the impact of the Internet can be seen as another example in a long history of technological transfers that promise economic regeneration for poorer regions, from the industries of railroads and textiles in the nineteenth century, to the production of South Korean automobiles, Japanese Walkmans, and Taiwanese silicon chips in the twentieth century. But can these hopes be fulfilled?

Multiple factors may potentially help to explain the global divide in Internet access, as in other forms of technological diffusion, and this study has been able to focus only on a limited range of explanations. Other plausible factors that should be considered in subsequent research include the role of cultural attitudes toward science and technology, the impact of public policy initiatives in education and training, as well as interest in the available contents of materials on the Web. The structures of opportunities on the Internet, as well as the resources of particular sections of the population, are likely to prove important reasons explaining patterns of use. We are only starting to explore these cross-national differences systematically and we need to go beyond the limitations of the available national-level aggregate data to understand the reasons why, for example, far more people are online in Finland rather than France, in Malaysia rather than in Vietnam, or in South Africa rather than Sudan.

Nevertheless the results of this analysis strongly suggest that the basic reasons for the global divide in Internet access are not strongly related to the particular characteristics of this type of information technology. The problem, it appears, is less whether Namibians lack keyboard skills, whether Brazilians find that few websites are available in Portuguese, or whether Bangladesh lacks network connections. Instead, the problems of Internet access are common to the problems of access to other communication and information technologies that have been widely available for decades in the West. Many of the poor living in Namibia, Brazil, and Bangladesh lack a rich information environment in terms of newspapers, radios, and television, as well as networked computers. In the first decade, the availability of the Internet has therefore reinforced existing economic inequalities, rather than overcoming or transforming them. The reasons are that levels of economic development combined with investments in research and development go a long way toward explaining those countries at the forefront of the Internet revolution and those lagging far, far behind. The regression models successfully explain a high level of variance. If countries have the income and affluence then often (but not always) access to the Internet will follow, along with connectivity to telephones, radios, and televisions. There are some important exceptions to this generalization – both affluent countries that lack extensive Internet access such as Kuwait and Greece, and middle-income countries that have made great progress toward online connectivity, such as Slovakia and Estonia. Economic determinism does not rule our virtual fate, but worldwide, without effective public policy interventions, it goes a long way toward predicting it.

The situation may change within the next few decades. In the West, the costs of access are declining, as computers become more and more affordable for businesses and consumers. U.S. consumer prices for personal computers have dropped by 43 percent over the last two years, and the costs of information service have fallen by one-third during the same period, with slightly slower declines in the price of software and mobile phones.[47] If the costs of Internet communications plummet further in future decades – for example, if Web access becomes widely available via throwaway prepaid mobile phones, like discardable Kodak cameras, connected to inexpensive satellite services circling the globe – then the gap between rich and poor could well change. In the current era, however, general levels of economic development and research investment go a long way toward predicting which countries have widespread access to the Information Society. Although the Internet remains in its adolescence as a new technology, we do not need to discard all our old explanations. Internet penetration rates can be predicted by economic models that also explain which countries are rich in telephones, radio and television, and even newspapers. Indeed, ownership of home computers in European households was strongly correlated with possession of many other prosaic consumer durables, including electric drills, video cameras, and clock radios, as well as ownership of a second automobile and vacation home.[48] Internet technology is new; global economic inequalities explaining technological diffusion are not. The next chapter goes on to explore whether we find a similar pattern that helps to explain the substantial differences in digital access evident between rich and poor within societies, even those at the forefront of the Information Revolution.

CHAPTER 4

Social Inequalities

For many the phrase "digital divide" has become a familiar catch phrase signifying the gap between information haves and have-nots, including splits along racial, gender, and class lines. The U.S. Department of Commerce has drawn attention to these disparities in successive studies since 1993. *Falling through the Net* emphasizes the lack of access to computers and the Internet commonly found in America among poorer households, those with only high-school education, the black and Hispanic populations, rural communities and women.[1] Pew surveys in spring 2000 confirm the familiar pattern found in many American studies, with sharp inequalities of Internet access by age, education, race, and ethnicity, plus the more modest gender gap (see Figure 4.1). Three-fourths of all American college graduates use the Internet compared with fewer than one-fifth of those who failed to graduate from high school. One-half of all whites are online compared with one-third of all blacks. And two-thirds of the younger generation is online compared with one in ten senior citizens. The OECD has documented similar patterns of stratification among the Internet population in Canada, Australia, and Finland.[2]

Why does this matter? The chief concern about the digital divide is that the underclass of info-poor may become further marginalized in societies where basic computer skills are becoming essential for economic success and personal advancement, entry to good career and educational opportunities, full access to social networks, and opportunities for civic engagement. Recognizing the potential problems of a two-track knowledge economy, the EU has prioritized the need for social inclusion as one of its objectives when launching the e-Europe Action Plan. The EU's strategy involves a combination of market and state initiatives, including reducing the costs of Internet access through

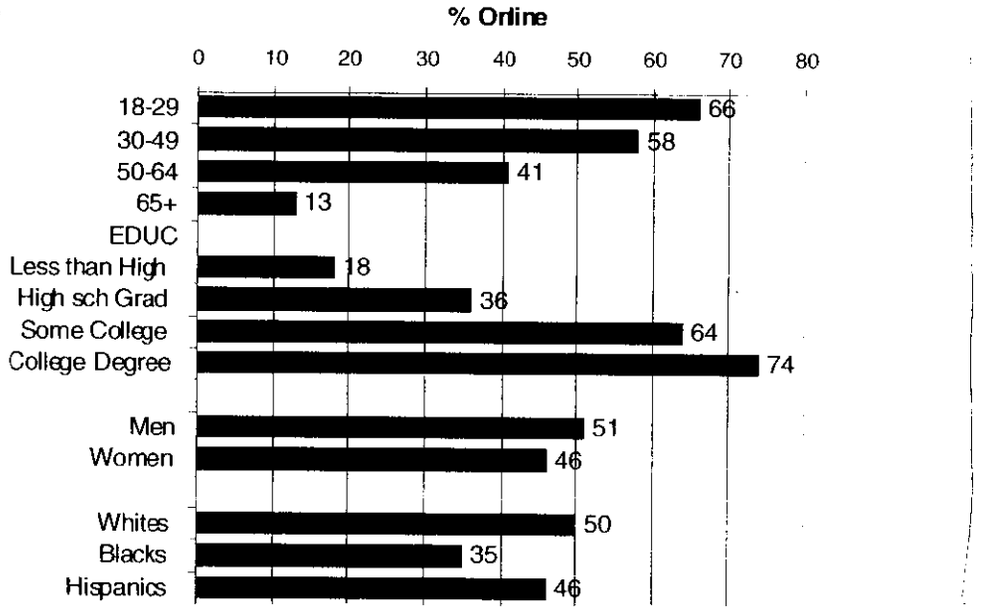


Figure 4.1. Percent Online by Social Group, U.S. 2000.
Source: Pew Surveys, United States, spring 2000.

telecommunications liberalization as well as making terminals and on-site training available in public spaces like unemployment offices, libraries, and schools.[3] Concern about Internet access has driven initiatives taken by the Blair government. The British Department of Trade and Industry has established a network of city learning centers and distributed reconditioned computers to homes in poor neighborhoods.[4] Similar public sector and nonprofit policy initiatives designed to overcome the digital divide have been launched recently in many other European countries. To evaluate whether these initiatives are likely to work we need to understand the underlying reasons for the digital divide and the likely projection of future trends. This chapter therefore focuses on three core questions:

- What theories provide plausible accounts of Internet diffusion?
- Who is online in postindustrial societies, and, in particular, does the social profile of Internet users found in the United States reflect that found elsewhere?

- Do the social inequalities in the online population reflect the particular characteristics of Internet diffusion per se, or does this pattern reflect disparities evident in access to older media such as cable television and fax machines?

THEORIES OF SOCIAL STRATIFICATION IN THE NETWORKED WORLD

Although attracting widespread concern, interpretations of the underlying reason for the digital divide remain under debate – for example, the appropriate weight that should be given to race and income in predicting American patterns of access and use. Even greater uncertainty surrounds the projection of future population trends on the Net and two mainstream perspectives have become common. On the one hand, the more optimistic *normalization* thesis discussed in Chapter 1 predicts that, at least in affluent postindustrial societies, the social profile of the online community will gradually broaden over time, like the early audience for radio or television, until eventually it mirrors society as a whole.[5] In countries at the forefront of the information revolution, such as the Unites States, Australia, and Britain, Internet use has rippled out within the last decade from an information source networking scholars and scientists at elite research institutions to become a medium of mass communications for the delivery of news, music, video and audio programming, as well as e-commerce and home shopping. The normalization thesis suggests that in developed societies the Internet could eventually become as popular as television today, so that we can expect penetration levels to reach 90 to 95 percent of the population, encouraged by falling costs for hardware, software, and services, and the growth of online home entertainment given fat-pipe delivery and technological convergence. Popular commentators swept up in the giddy hype of new gizmos and gadgets, surrounded at home and work by easy access to all Internet, all the time, often assume that eventually most will succumb to the digital onslaught. Competition in the marketplace, some assume, will eventually correct any residual major disparities, removing the need for government intervention or regulation for the provision of universal service, beyond the minimal prevention of monopolistic practices.[6]

In contrast the more pessimistic *diffusion* theory developed by Everett Rogers provides an alternative interpretation of social stratification in technological adaptation.[7] We can draw on case studies of the impact of technological innovations ranging from the invention of stirrups and gunpowder in military strategy, to the telegraph, railway, and steam engine in the nineteenth century, and airplanes, automobiles, or telephones in the twentieth. Research has produced hundreds of case studies from the spread of new crops studied by rural sociologists to the implementation of new teaching techniques studied by educational psychologists. These studies have commonly established that, compared with laggards, early adopters of new innovations are characteristically drawn from groups with higher socioeconomic status. Education, literacy, and social status provide access to the essential financial and information resources required to adapt flexibly to innovative technologies. Moreover, diffusion theory suggests that the adoption of successful new technologies often reinforces economic advantages, like the greater productivity facilitated in agriculture by mechanized tractors, artificial fertilizers, and specialist seed-corn, so that the rich get richer, and the less well-off sectors fall farther behind. Everett Rogers emphasizes that this pattern is far from inevitable, because the conditions under which an innovations are implemented determine, in part, their social consequences. Active initiatives to level the playing field by the state and nonprofit sectors can broaden technological access. The existing social structure also plays a role; innovations in highly stratified societies will usually reinforce existing socioeconomic disparities. The type of technology can also influence this process, for example, the initial resources required for access, including both financial investments and educational skills. Nevertheless diffusion theory predicts that without successful state intervention, if the spread of the Internet follows the conventional trajectory established by many previous technologies, then the initial adoption of wired computers can be expected to exacerbate existing social divisions, at least in the early-to-middle stages of the S-shaped diffusion curve, and perhaps in the longer-term as well if the new technology produces substantial productivity gains and continuing access barriers.

The cross-national evidence on this issue remains scattered and inconclusive, but some indicators within the U.S. Internet population lend some weight to support the normalization thesis. After analyzing trends from 1995 to 1998 in Harris national surveys, David Birdsell and colleagues concluded that as the proportion of U.S. users surged from 14 to 58 percent of all Americans, the demographics of the online community broadened: "Once heavily overbalanced by male users, the Web is now accessed by men and women almost equally. And once predominantly white, the Web population now reflects a racial breakdown statis-

tically indistinguishable from Census data for the general population... The Web reflects America much more accurately today than when the technology was in its infancy."[8] Similarly, the 1998 Pew Research Center study, *The Internet News Audience Goes Ordinary,* also reported greater diversification in the American online community: "Increasingly people without college training, those with modest incomes, and women are joining the ranks of Internet users, who not long ago were largely well-educated, affluent men."[9] Other research reports that the gender gap in the U.S. Internet population has become insignificant in recent years.[10] A Stanford Institute's study by Nie and Erbring suggests that racial differences in online access in America have become less important today than an income differentials, conclusions echoed by the Forrester Report.[11]

However, the matter has not been settled because other American evidence points in the opposite direction. The 1999 report *Falling through the Net* emphasized that in terms of racial, educational, and income inequalities the digital divide between those with access to new technologies and those without widened from the mid- to late-1990s, not narrowed. The study by the U.S. Department of Commerce concluded that ethnic differences in the virtual world cannot be accounted for solely by levels of affluence since in their adoption of home computers and links to the Web, African Americans lag substantially behind white Americans within every income category. "A White, two-parent household earning less than $35,000 is nearly three times as likely to have Internet access as a comparable Black household and nearly four times as likely to have Internet access as Hispanic households in the same income category."[12] Anthony Wilhelm confirmed that racial and ethnic differences in computer usage have not disappeared, since his analysis of the 1994 U.S. Current Population Survey data found that these differences persist even after controlling for education and household income.[13] A detailed study of trends in computer ownership and Internet use from 1997 to 1998 by Hoffman and Novak also concluded that the overall gap between whites and African Americans increased from the mid- to late-1990s.[14] Moreover it remains uncertain whether "normalization" of African-American participation in the online population will eventually occur even if use of the Web eventually reaches 90 to 95 percent of all Americans, because there are racial disparities in access to far more basic and long-established technologies, such as household telephones.

To clarify the broader pattern, and to generalize more widely beyond America, survey evidence needs to compare Internet trends in a wide

range of societies at different levels of development. The most straightforward approach to testing these theories would be to compare time-series data to observe whether the digital divide has widened or narrowed during the last decade. Yet benchmark survey data during this period remain limited, even in the United States. Given the rapid pace of technological change, which seems likely to continue, it remains difficult to project future trends from the social characteristics of early adopters. The available evidence on the digital divide in the United States may also reflect endemic inequalities within this particular society, such as the deep-rooted cleavage of race, rather than general patterns characteristic of those who participate in the online community in more egalitarian societies, such as the smaller welfare state societies of Sweden and the Netherlands. Even fewer surveys have monitored trends in Internet use across many different countries, although more information is rapidly becoming available from market research companies, generated by the demands of e-commerce.[15] For comparative data this chapter draws on the biannual series of Eurobarometer surveys conducted since 1970 in the 15 EU member states by the European Commission.[16] For the United States we analyzed successive surveys conducted on a regular basis since 1995 by the Pew Center on the People and the Press, providing perhaps the most detailed continuous time-series data on Internet use available in America. The survey items are not identical in both series but functionally equivalent items allow comparison of demographic patterns of Internet use from spring 1996 to 1999. The main conclusions were cross-checked against secondary sources for comparison with other nations.

The comparison of the United States and Europe includes both leaders and laggard societies. This allows us to test the "normalization" claim that the size of the digital divide relates to the level of Internet diffusion. The fifteen member states in the European Union provide a range of postindustrial societies that are relatively similar in their levels of socioeconomic development, political systems, and cultural traditions. All are established democratic states, with advanced economies and extensive welfare services. This comparison follows the classic logic of the "most similar system" design, which assumes that the factors common to relatively homogeneous types of society are irrelevant to explaining their differences.[17] Roughly equivalent levels of literacy, education, and affluence within European nations, despite the existence of poorer regions, means that these factors can be largely discounted in seeking explanations of the digital divide. The social structure remains

Access Reading Rooms

Table 4.1. *Trends in Internet use in Europe and America, 1996–1999*

	Spring 1996	Spring 1997	Fall 1998	Spring 1999	Increase 1996–1999
Sweden	12	26	43	61	+49
United States[a]	21	36	42	49	+28
Denmark	10	17	26	44	+34
Finland	11	16	18	39	+28
Netherlands	9	16	19	32	+23
Luxembourg	5	13	16	22	+17
Britain	9	10	11	22	+13
Northern Ireland	4	8	10	20	+16
Italy	3	5	7	14	+11
Ireland	4	5	9	14	+10
Austria	4	10	7	11	+7
Belgium	3	6	8	11	+8
France	2	4	4	9	+7
Germany West	5	8	8	8	+3
Germany East	2	4	5	8	+6
Spain	2	2	5	8	+6
Greece	1	3	3	7	+6
Portugal	2	2	3	5	+3
EU15	***5***	***9***	***12***	***20***	***+15***

[a] United States: successive surveys by The Pew Research Center for the People and the Press. See *www.peoplepress.org.*

Notes: The Eurobarometer question asks, *"Do you have access to, or do you use, the Internet or World Wide Web."*[33] The Pew survey asks, *"Do you ever go online to access the Internet or World Wide Web or to send and receive email?"*

Sources: Eurobarometer 44.2, spring 1996; 47.0, spring 1997; 50.1, fall 1998; and 51.0, spring 1999.

relatively similar, in terms of the predominance of class cleavages, although with a complex pattern in terms of ethnicity and race.[18] Moves toward removing tariff barriers, economic and monetary union, and harmonization of economic and social policy within the European Union also mean that this region shares relatively similar telecommunications policies, with market liberalization evident in most member states during the last decade. At the same time, and this is most important, the European nations under comparison vary greatly in terms of the dependent variable that we are seeking to explain, namely levels of

Table 4.2. *Trends in computer use in Europe and America, 1996–1999*

	Spring 1996	Fall 1997	Spring 1999	Increase
Sweden	43	62	73	+30
United States[a]	60	66	69	+9
Denmark	49	61	65	+16
Netherlands	54	61	64	+10
Finland	36	43	52	+16
Luxembourg	41	49	48	+7
Britain	41	47	45	+4
Northern Ireland	25	34	39	+14
Italy	31	32	37	+6
Belgium	28	32	37	+9
Austria	23	41	33	+10
Spain	25	29	33	+8
Ireland	23	27	31	+8
France	25	34	30	+5
Germany West	31	32	29	−2
Germany East	27	32	27	0
Portugal	21	20	22	+1
Greece	12	19	17	+5
EU15	***31***	***38***	***40***	***+9***

[a] United States: successive surveys by *The Pew Research Center for the People and the Press.* See *www.peoplepress.org*

Note: The Eurobarometer question asks, *"Do you have access to, or do you use, a computer?"* [34] The Pew surveys ask: *"Do you use a computer at your workplace, at school or at home on at least an occasional basis?"*

Sources: Eurobarometer 44.2, spring 1996; 47.0, spring 1997; 50.1, fall 1998; and 51.0, spring 1999.

Internet connectivity. Overall one-fifth of the European population is online, but Nordic nations have forged ahead to the forefront in the digital age while Mediterranean societies often lag behind the European average (see Tables 4.1, 4.2, and Figure 4.2). In spring 1999, almost two-thirds (61 percent) of all Swedes were online. Sweden had almost ten times as many Internet users than Portugal and Greece. Use of personal computers confirms parallel contrasts, with use ranging from three-fourths of the population in Sweden and two-thirds in Denmark and the Netherlands down to less than one-fifth in Greece. Similar cross-national patterns are evident with the distribution of Internet hosts

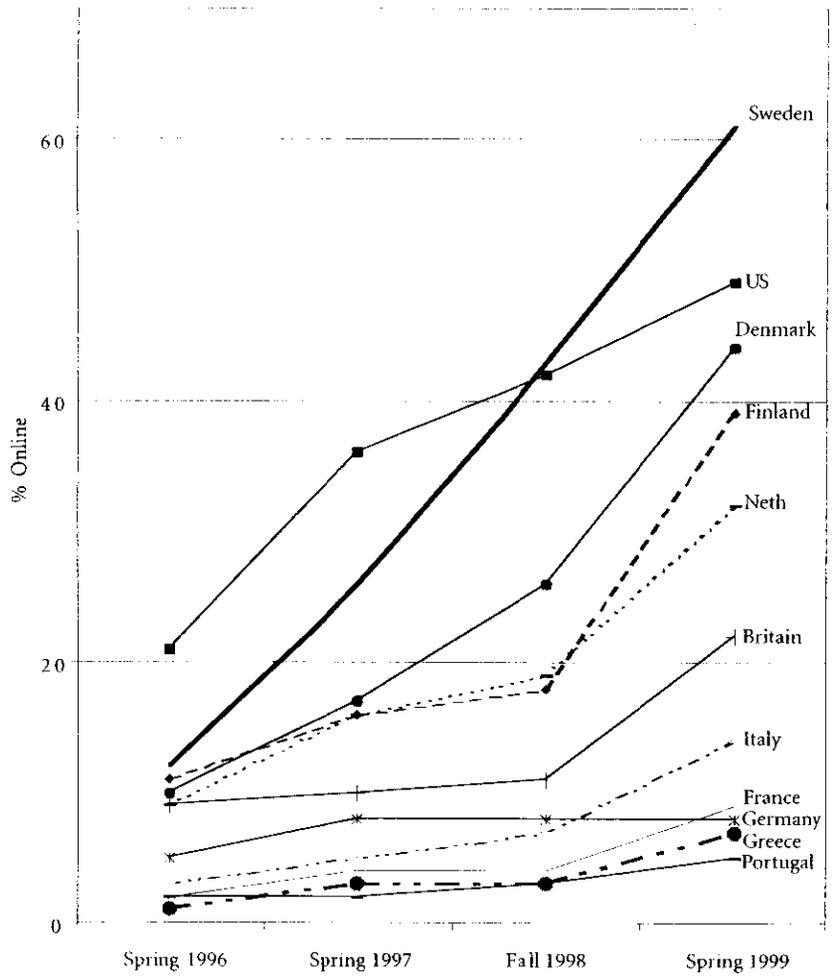


Figure 4.2. Percent Online, EU-15, 1996–99.
Sources: Eurobarometers 1996–1999; Pew Surveys, United States.

across Europe. If the "normalization" predictions are correct, we might expect to find that the *relative* size of the digital divide by social status, age, and gender would diminish most in societies at the forefront of the Information Society, including Sweden and Denmark. The social and demographic characteristics of the online community in different countries can be described first before multivariate regression analysis is used to examine the relationship among these variables.

WHO IS ONLINE?

INCOME

The digital divide is a multidimensional phenomenon tapping many social cleavages but differences of resources are commonly assumed to be among the most important, meaning the capacities based primarily on the income, occupation, and education that people bring to using new forms of info-tech. *Falling through the Net* emphasized that household income was one of the strongest predictors of Internet access in America.[19] U.S. Census data show that home ownership of PCs quadrupled from 1984 to 1997, but this period saw growing disparities in ownership among social strata based on household income, race, and education.[20] An OECD study, drawing on data from France, Japan, and the United States, confirmed the substantial disparity in the availability of personal computers in the home for different levels of household income, with the size of the gap between the lowest and highest income groups widening from 1995 to 1998.[21] Economic resources, including personal or household income, influence the ability to afford home computers and modems, related software, and the monthly ISP and telephone or broadband cable connection charges. Telephone costs can be substantial in Europe, especially where charges for local calls are metered by the minute, with these bills outweighing the initial investment in computer hardware within a few years. The OECD's *Information Technology Outlook 2000* study found that the growth in Internet demand has been driven by a combination of faster connection speeds, improved reliability and service, easier technical use, and declining access costs. Dial-up telephone modems currently remain the most popular mode of household access, used in two-thirds of all homes in OECD member states, although more advanced forms of delivery are gradually becoming more available, including cable, DSL, ISDN, and wireless.[22]

How far does household income determine Internet access? Table 4.3 summarizes trends in the social profile of Internet users within Europe from 1996 to 1999, for comparison across all major social cleavages. Figure 4.3 illustrates the size of the gap in Internet access between the most and least affluent quartile household incomes, with the nations ranked across the chart by their overall level of Internet penetration. Three major findings stand out. First, as expected, the income gap in Internet users across the whole of Europe proved sub-

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Table 4.3. *Trends in the social profile of Internet users in Europe, 1996–99*

	Percent online spring 1996	Percent online spring 1999	Change
All EU-15	5	20	+15
Age			
15–25	9	32	+23
26–44	7	24	+17
45–64	5	16	+11
65+	1	3	+2
HH income category			
–	4	14	+10
–	3	14	+11
–	5	22	+17
++	10	37	+27
Age finished education			
Up to 15	1	5	+4
16–19 years	4	15	+11
20+	9	33	+24
Gender			
Men	6	22	+16
Women	4	17	+13
Occupational status			
Managers	14	44	+30
Other white collar	8	29	+21
Manual worker	3	15	+12
Home worker	2	8	+6
Unemployed	3	10	+7
Student	13	44	+31

Sources: Eurobarometer 44.2, spring 1996; 47.0, spring 1997; 50.1, fall 1998; and 51.0, spring 1999

stantial; on average the wealthiest European households were almost three times more likely to be online than the poorest ones. Overall 37 percent, of those living in the most affluent households were online, compared with only 14 percent, of those in the poorest homes. There was a consistent and significant association between household income and levels of Internet access across all EU countries, with the single exception of Greece.[23] Moreover, across Europe the relative size

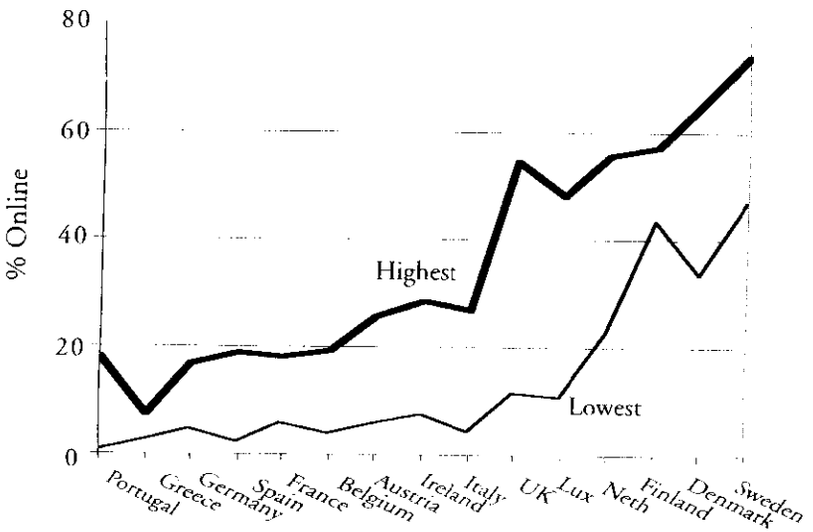


Figure 4.3. Percent Online by Household Income, EU-15, 1999.
Source: Eurobarometer 51.0, spring 1999.

of the gap between rich and poor stayed roughly constant from spring 1996 to spring 1999, rather than increasing or diminishing. During these years the EU Internet population grew at a rate of roughly 10 percent per annum but this did not lead to any closure of the digital divide by income. Finally, the comparison of societies that are leaders and laggards in the Information Age gives no support to the normalization thesis claim that income differentials necessarily diminish as Internet use widens throughout the population; if anything the reverse. Despite relatively widespread use of the Internet in Britain, for example, the most affluent households were five times more likely to be online than the poorest. The variations among European countries suggest that many factors within each nation may influence the income gap in Internet access, such as state initiatives to make wired computers widely available through community centers, unemployment offices, and schools, as well as levels of market competition driving down the financial costs of hardware, software, and access charges. But the differential between rich and poor families evident in such countries as Britain, Luxembourg, and Denmark mean that, at least during the emerging years, we would not expect the income gap to close automatically as the Internet diffuses more widely throughout society.

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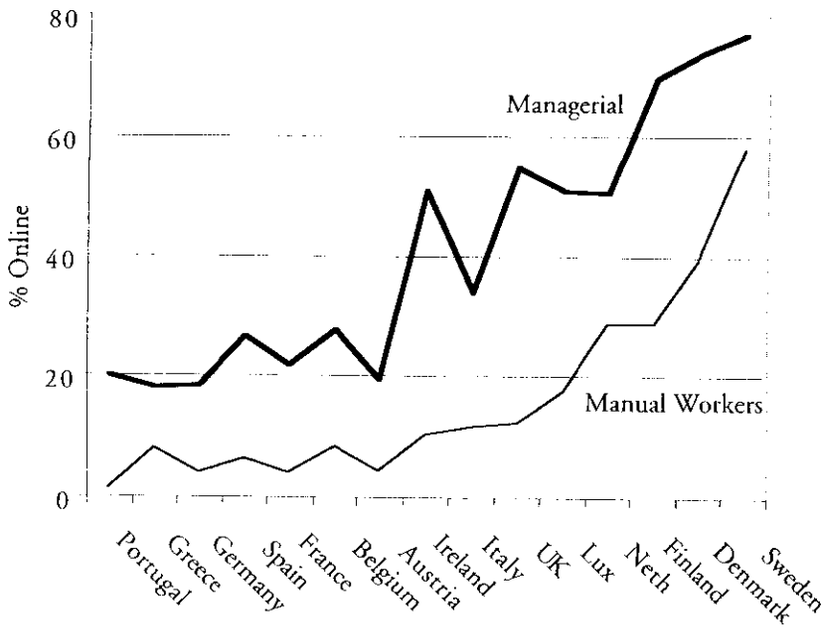


Figure 4.4. Percent Online by Occupation, EU-15 1999.
Source: Eurobarometer 51.0, spring 1999.

OCCUPATION

Related patterns in the workforce can also be expected to be important for many reasons. Professional and managerial jobs in the service sector facilitate 24/7 Internet connections at the office, often through high-speed LAN networks, as well as providing training assistance and technical support. Companies also commonly provide managers and executives with mobile equipment such as laptops, digital personal assistants, and cell phones, as well as subsidizing home access charges, to facilitate connectivity with the office. Professional, managerial, and executive salaries provide the resources to afford consumer durables like computers and high-speed cable connections for the home and family, and multiple-connection household intranets. In contrast, manual workers, although using computers as part of the industrial manufacturing process, are less likely to experience Internet access at work or to acquire the skills and experience in the workplace that breed comfort and familiarity with the Web at home. Governments in Britain, Germany, and Sweden have emphasized the need to bring the unemployed into the knowledge economy, through the provi-

sion of networked computers and training in job centers and unemployment offices. Nevertheless, even with these initiatives, those seeking work are likely to be among the most marginalized and poorest members of society. There are reasons to believe that occupational status may prove less important for Internet access as use diffuses more widely; during the last five years home access has doubled in America, with remarkably little increase in the proportion using the Internet from work.[24]

Figure 4.4 shows the distribution of Internet access by the respondent's occupation in the workforce in the fifteen member states of the EU. The pattern confirms that managers and professionals are almost twice as likely to use the Internet as those in other white-collar jobs including clerical assistants and service sector employees, and managers are almost three times as likely to use the Internet as manual workers. Access for the unemployed fell just below the level of manual workers. Again, the gap between the info-rich and poor varied across nations but it proved largest among certain leader societies including the United Kingdom, Finland, and Denmark, and slightly more marked than the disparities already observed by household income.

EDUCATION

The related divide in Internet use by educational attainment is well established in many American studies, where college graduates are among those most familiar with the Internet. Wilhelm, for example, concluded that education was a stronger determinant of connectivity in America than any other demographic or social variable.[25] Many reasons can be given for this pattern. Schools and colleges provide an environment that is exceptionally rich in all forms of info-tech and indeed these have usually been among the first institutions wired to the Net in most countries. Education can be expected to improve the general capacity for analytical reasoning and information filtering, which helps cope with the flow of information available online, as well as strengthening numeracy, literacy, English-language, and keyboard skills. Schools and colleges commonly provide students with free email and Web hosting facilities, computer labs, as well as direct hands-on experience of surfing the Internet for research, and training support or technical backup for using common software packages. College education is also closely related to subsequent occupational status and income, which we have already observed are important, as well as being linked to generational patterns of diffusion, illustrated by the popularity of programs for music sharing such as Napster on college campuses. Figure 4.5 illus-

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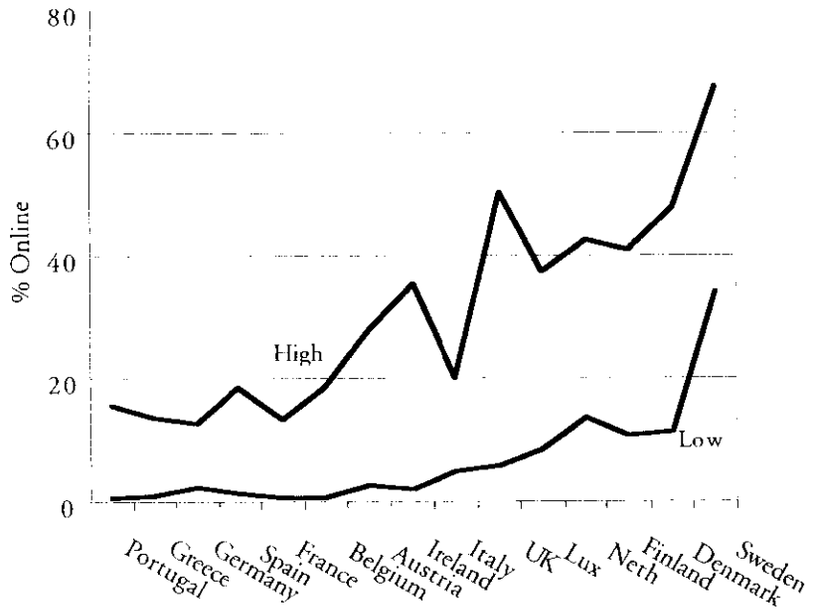


Figure 4.5. Percent Online by Education, EU-15 1999.
Source: Eurobarometer 51, spring 1999.

trates the relationship between the age of finishing education and use of the Internet in European member states. The results confirm the expected disparities by education, showing similar results to the inequalities evident in the workforce. In Europe, those with a college education are seven times more likely to be online than those who left school at 15. Indeed education proved one of the strongest predictors of connectivity: more than 40 percent of college students in Europe are online, a figure ranging up to 80 percent of all students in Sweden and Finland.

GENDER

The gender gap within the online community has been the subject of widespread study. Some surveys have reported that this difference has closed recently in America; for example the Pew Internet and American Life tracking survey suggested that by spring 2000 the surge in the number of women online had produced gender parity, although women and men continued to differ in their attitudes toward new technology and in their behavior online.[26] Nevertheless the evidence about the gender gap remains inconclusive; for example, AC Nielsen's *Net Watch* surveyed thir-

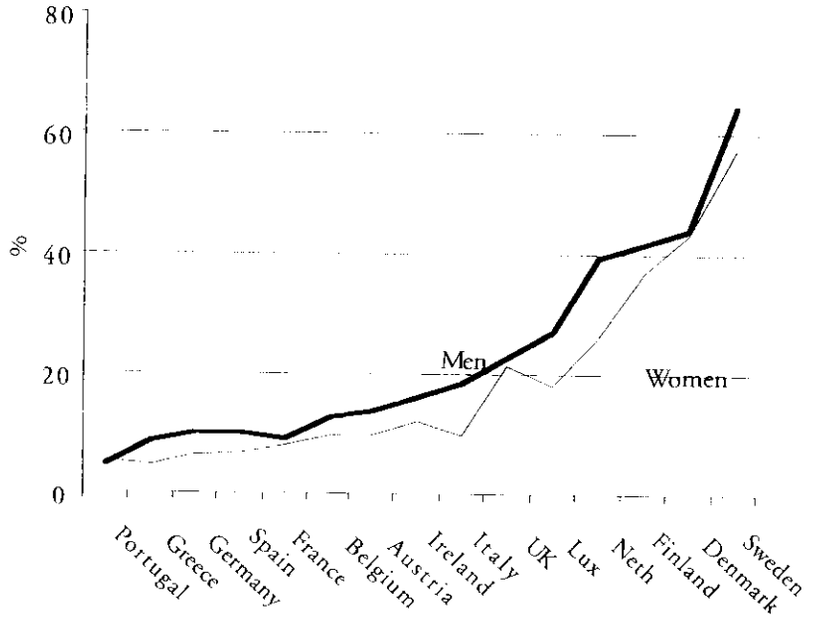


Figure 4.6. Percent Online by Gender, EU-15, 1999.
Source: Eurobarometer 51.0, spring 1999.

teen nations in North America, Europe, and Asia in spring 2000, and reported that women were less likely to be online in every country, including the United States, with almost twice as many male to female Internet users in Germany, Hong Kong, and Taiwan.[27] Many plausible reasons may account for any gender differences in use of computer technology and within the online community. Bolt and Crawford reviewed a wide range of evidence suggesting that girls and women are less likely to use computers because of their early experiences within school classrooms, reflecting long-standing gender differences in attitudes toward science and technology, as well as the typical contents of computer games and websites available for children.[28] The position of women as primary caregivers in the home and family may also play a role, since we have already observed the importance of work environments for Internet access. In Europe, the evidence on male and females within the online community in Figure 4.6 shows that the gender gap proves the weakest predictor of Internet use among all the factors we have considered so far, with the difference between women and men becoming statistically insignificant in Belgium, Denmark, France, Portugal, the United

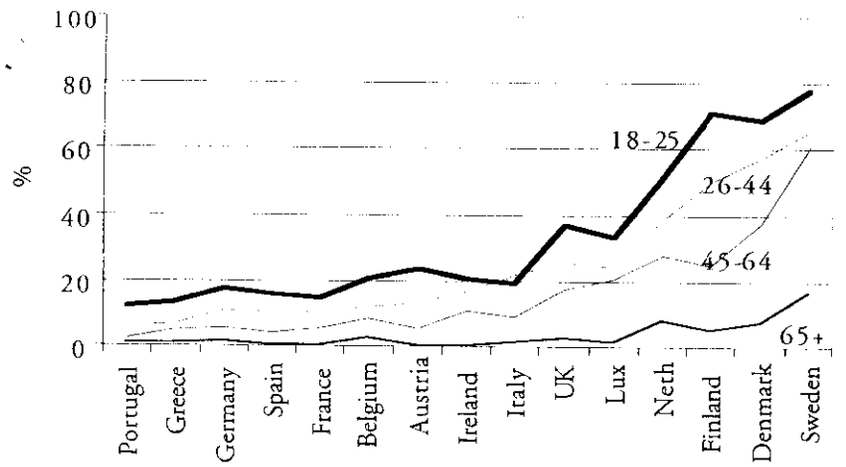


Figure 4.7. Percent Online by Age Group, EU-15, 1999.
Source: Eurobarometer 50.1.

Kingdom, and Finland. Nevertheless a significant online gender gap was evident elsewhere, especially in the Netherlands, Sweden, and Italy.

GENERATION

The generational difference in adaptation to new technologies is perhaps the most significant for the future diffusion of the Internet, and yet the most taken for granted in policy circles. The pattern in Figure 4.7 shows that early adopters in Europe were concentrated among the youngest age groups, with minimal use in most of Europe among senior citizens – this despite the fact that the Internet seems well suited to the needs of the elderly, as a fairly sedentary population with considerable leisure time, especially for social networking, hobbies, and services such as the home delivery of groceries. The typical age profile may flatten in future because there is a large potential market among the elderly if access to the Internet becomes more commonly delivered through dedicated plug-and-play email units, and services like Web-TV, rather than via more technically demanding computers. At present, the online community in Europe largely excludes the retired population. The generation gap within the Nordic region, where the Internet has penetrated most widely, proves the largest of any social cleavage observed so far. The youngest group is ten times as likely to be online than the oldest group; overall almost a one-third of all Europeans under 25 are online, compared with only 3 percent of the over-65 year-olds. The Napster

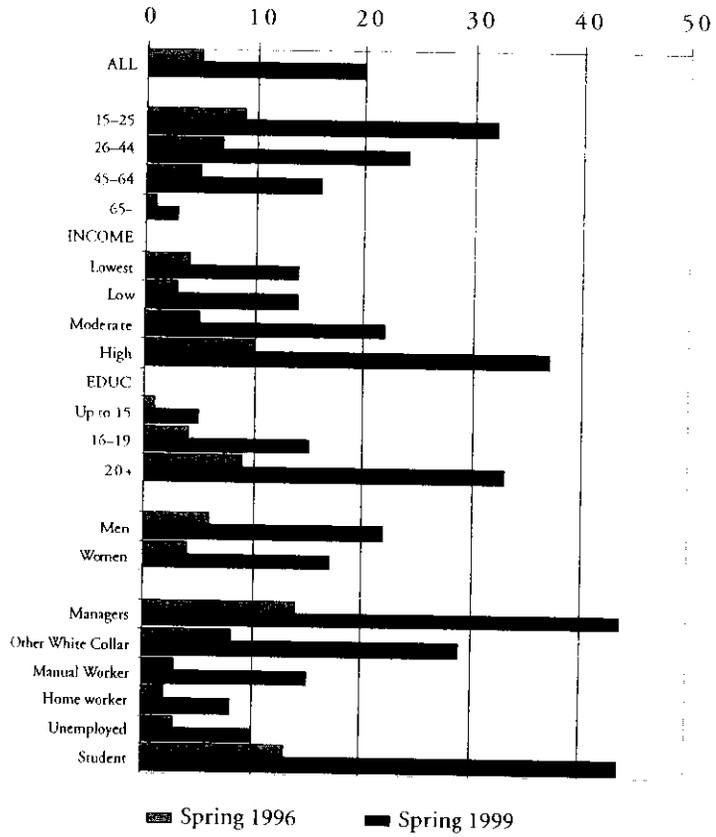


Figure 4.8. Change in the Percent Online by Social Group, EU-15.
Source: Eurobarometers 1996 and 1999.

generation is already experiencing a virtual world as they develop that is different from formative lives of their parents and grandparents.

The evidence so far suggests that the familiar social profile of online users in the United States is also evident throughout Western Europe. Trends in these societies since the mid-1990s confirm that, contrary to the "normalization" thesis, the resource-based inequalities that we have already observed grew in significance as Internet use gradually diffused more widely (Figure 4.8). Multivariate analysis predicting use of the Internet in spring 1996 and 1999 helps to distinguish the relative weight of the factors discussed so far and confirms the significance of these patterns. The order of the variables in the OLS regression models fol-

lowed the standard logic of the classic "funnel of causality".[29] The demographic variables of age and gender were entered first, followed by the resource variables of household income, education, and the respondent's occupation, the latter coded as manual or nonmanual. The country variables were then entered as dummy variables (0/1) to determine if the national differences in Internet use remain significant even after controlling for social background.

The results of the models confirm that all the demographic factors discussed in this chapter were significantly associated with Internet use in Europe (see Table 4.3). In particular, the gender and age differences remained significant predictors of Internet use even after controlling for income, education, and occupation, suggesting that resources are only one part of the explanation here. Moreover the national-level country variables usually continued to be significant, even after controlling for social background. This indicates that the structure of opportunities within each nation (discussed in the previous chapter) influences Internet use even after controlling for individual-level differences in occupation and income. For example, a well-paid college-educated company manager living in Stockholm or Copenhagen would still be more likely to surf the Web or use email than an equivalent colleague working in Madrid or Athens. The macrolevel context is important, including such factors as state initiatives to widen access through education and training, and market competition in the pricing of computer hardware and ISP connectivity.

Turning to the analysis of changes over time, the comparison of the same models run in 1996 and 1999 demonstrate that, far from equalizing, the digital divide in Europe expanded during these years; the inequalities of access by income, education, occupational status, and age become stronger and only the gender gap weakened over time. It is likely that the social profile of users may possibly flatten further within the next decade, if Internet access spreads even more widely to become ubiquitous in almost all European households, as commonplace as a VCR or refrigerator. Nothing that we have said disproves the claims that the "normalization" thesis could still be true in the longer-term. But the European evidence indicates a growing digital divide between the information-rich and poor during the emergent Internet era, in addition to the widening North–South global disparities documented in the previous chapter, with no evidence to date that these gaps are starting to close or normalize in leading societies where use of the Internet has become most pervasive.

RELATIVE INEQUALITIES IN THE INFORMATION SOCIETY

The fact that there are absolute differences in access to the Internet is hardly surprising. However, the question remains whether the *relative* disparities in access to the Internet are substantially different or similar to the distribution of other common forms of information and communication technology, such as VCRs and cable TV. Like gambling at Rick's bar, some popular accounts are shocked – *shocked* – to discover social inequalities on the Internet. We should not be. As in the previous chapter, the more realistic approach compares relative inequalities in computers access and Internet use with disparities in the distribution of other common forms of information technologies. The spring 1999 Eurobarometer survey monitored whether people had access to or used ten different types of information technology, including the Internet, as well as VCRs, facsimile machines, and personal computers. If we establish that patterns of income, educational, or occupational inequalities are similar across all types of communication and information technologies, then this suggests broad explanations of this phenomenon relating to deep-rooted patterns of social stratification endemic in modern societies. Households with Internet access can be expected to possess multiple consumer durables for entertainment and communications, such as satellite or cable TVs, pagers and fax machines, home entertainment centers, and mobile phones. Conversely, if the distribution of Internet access differs from use of other types of info-tech, then we should search for explanations based on the distinctive characteristics of the Internet per se, such as the financial costs of gaining online access, the cognitive skills and computing experience required for surfing, language barriers to use of the Web in non–English-speaking countries, and the way people respond to the type of materials and services available online (e.g., music, entertainment, and shopping). Both relative and absolute inequalities can be regarded as equally important, but the analysis of the former provides deeper insights into the causes of this phenomenon and therefore also its potential solution.

The correlations shown in Table 4.4 indicate that Internet use is significantly associated with access to all forms of communication technologies, including noncomputer related machines such as VCRs and cable TV. Reflecting parallel patterns at national level documented in the previous chapter, this suggests that individuals living in affluent households with many different forms of consumer durables designed for traditional forms of home entertainment and communication are also

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Table 4.4. *Explaining Internet use in Europe, 1996 and 1999*

	1996			1999		
	R	Sig.	B	*R*	Sig.	B
Demographics						
Age	–.085	.000	–.035	–.168	.000	–.642
Gender	.058	.000	.588	.052	.000	.327
Resources						
Education	.095	.000	.303	.153	.000	.609
Income	.055	.000	.783	.141	.000	.439
Class	.056	.000	.827	.077	.000	.574
Nation						
Greece	–.035	.000	–1.43	–.073	.000	–1.72
Germany	–.006	.849	–.035	–.067	.000	–1.30
France	–.037	.000	–.774	–.057	.000	–1.24
Spain	–.027	.000	–1.03	–.055	.000	–1.36
Portugal	–.007	.012	–.563	–.054	.000	–1.44
Belgium	–.015	.000	–.628	–.052	.000	–1.20
Austria	–.001	.923	.020	–.041	.000	–.93
Italy	–.018	.000	–.507	–.036	.000	–.91
Ireland	.010	.104	.359	–.029	.000	–.73
United Kingdom	.069	.000	.966	.000	.432	.23
Netherlands	.035	.000	.578	.021	.000	.45
Finland	.055	.000	.784	.035	.000	.65
Denmark	.038	.000	.573	.049	.000	.89
Sweden	.068	.000		.099	.000	1.77
Cox-Snell R2	.073			.278		
Nagelkerke R2	.209			.431		
Percent correct	94.5			83.8		

Notes: The table reports the coefficients predicting the use of the Internet based on binary logistic regression models. Use of the Internet is measured as a dichotomy where 1 = yes and 0 = no. Luxembourg, which is close to the Europe mean, is excluded from the standardized household income scale. Age in measured in years. Education is measured as years finished full-time education. Income is measured from the standardized household income scale. Class is based on the manual/nonmanual occupation for the head of household. Gender is 1 = male and 0 = female.

Source: Eurobarometer 47, spring 1996 (No. 65178) and spring 1999.

Table 4.5. *Use of new and old media in Europe, 1999*

	Inter-net	Com-puter	CD ROM	Modem	VCR	Fax	Sat. TV	Cable TV	Video-text
Computer	**0.61**								
CD-ROM	**0.59**	**0.70**							
Modem	**0.91**	**0.67**	**0.65**						
VCR	0.19	0.31	0.26	0.21					
Fax	0.44	0.45	0.43	0.47	0.22				
Satellite TV	0.12	0.14	0.13	0.12	0.12	0.13			
Cable TV	0.14	0.16	0.16	0.15	0.15	0.14	0.24		
Teletext TV	0.19	0.25	0.22	0.21	0.32	0.18	0.16	0.10	
Videotext	0.09	0.12	0.11	0.11	0.07	0.14	0.06	0.13	–0.02

Note: Q: "*Do you use, or do you have access to...*" The figures represent individual-level simple correlations (*R*) without any controls.

Source: Eurobarometer 51.0, spring 1999 EU-15.

most likely to access networked computers. As we saw in the previous chapter, ownership of personal computers is related to all sorts of other common household gadgets from deep-fat fryers to video cameras and clock radios. There can obviously be many exceptions; for example, less affluent students, low-paid service professionals, and office clerical workers are commonly in work environments where the Internet is easily available even if they lack home access. Nevertheless, the association between use of computer and other types of household consumer durables implies that broad and deep-rooted patterns of social stratification are the major explanation for patterns of Internet diffusion.

To test this further, multivariate models were run as before with three dependent variables, the percentage of the population using the Internet (% Online), a broader New Media index summarizing access to a computer, CD-ROM, and modem, as well as the Internet, and an Old Media index including access to traditional technologies, including a VCR, fax, satellite TV, cable TV, teletext TV, and videotext. Most strikingly, comparison of the results across all three models shows that similar social and demographic factors explaining online participation also help predict access to new *and* old media technologies (see Table 4.6). The digital divide is striking but far from new. The relative strength of the coefficients varied slightly across models; for example, education and occupation were slightly more strongly associated with the new

Table 4.6. *Explaining use of new and old media in Europe, 1999*

	Percent Online		New media		Old media	
	Beta	Sig.	Beta	Sig.	Beta	Sig.
Demographic						
Age (years)	–0.12**		–0.16**		–0.18**	
Gender (male)	0.06**		0.05**		0.07**	
Resources						
HH income	0.16**		0.21**		0.25**	
Education	0.12**		0.15**		0.06**	
Manual occupation	0.09**		0.12**		0.06**	
Region						
North (1) South (0)	0.23**		0.21**		0.25**	
Constant	0.21		0.81		2.19	
Adjusted R^2	0.19		0.26		0.24	

** Sig. > $p.$ = .01

Notes: The figures represent standardized beta coefficients using OLS regression models. **Percent Online:** Have access to or use the Internet or World Wide Web (0 = no/1 = yes). **New Media Index:** four-point scale measuring use or access to computer plus CD-ROM plus modem plus Internet. **Old Media Index:** six-point scale measuring use or access to VCR plus Fax plus Satellite TV plus Cable TV plus Teletext plus Videotext.

Source: Eurobarometer 51.0, spring 1999.

than old media. Nevertheless the headline finding concerns the striking similarities across models. This pattern confirms that important social inequalities exist in the virtual world. Since the mid-1990s, the global divide between leader and laggard nations, and the social divide among subpopulations even within leader nations, have expanded substantially. In Europe, as in the United States, the sweeping tide of the Internet has left behind many poorer households, manual workers, the less educated, the elderly, and women. Yet, there is nothing distinctive about these social and regional inequalities in the virtual world, which also characterize access to the Information Society delivered via older media technologies such as cable or satellite TV, VCRs, and fax machines. We may be less concerned about the implications of lack of access to cable TV or VCRs than lack of access to the Internet, but this

insight has important implications for policy initiatives designed to overcome the social barriers to digital access. The results suggest that such programs as training in keyboard skills or wiring schools may help to overcome the digital divide but they are likely to have limited effect given the deep-rooted socioeconomic barriers to access.

At the same time it is true that in the emerging knowledge economy the evidence of trends in Internet use remains limited, and we remain at the bottom of the diffusion curve. The situation could change in the near future and the social profile of the online community could be transformed if costs fall dramatically, driven by the proliferation of subscription-free and unmetered access services, greater competition, and the increasing availability of broadband services, as many predict.[30] Companies also forecast a proliferation of Web appliances delivering a cut-down version of the Internet outside the box. Although there are reasons to be skeptic about industry hype, eTForecast estimate that there are currently 21.5 million Web-enabled appliances in use globally, such as cell phones and personal assistants. The projected figures for 2005 are 596 million globally: 115.4 million for the United States and 126.4 million for Western Europe. Web delivery via mobile phones has proved particularly popular in Japan where home PCs are not common.[31] The popular distribution of powerful but relatively inexpensive units such as Play Station II also facilitates new ways of accessing the Internet. The consequences of these developments for the digital divide remain to be determined but their lower costs could help close the divide.[32]

CONCLUSIONS: SOCIAL STRATIFICATION AND INTERNET ACCESS

The explanation for the digital divide is often assumed to lie in certain characteristics of this technology, such as the need for computing skills and affordable online connections. The policy solutions designed to ameliorate the digital divide commonly focus on just these sorts of fixes, such as wiring schools and classrooms, training teachers, and providing community access in poorer neighborhoods. Certainly this can do no harm. But will these initiatives work in terms of diversifying the online population? The results of this analysis suggest that, unfortunately, it seems unlikely. The policy fixes are too specific, the problem of social inequalities too endemic.

The analysis in this chapter demonstrates that the heart of the problem lies in broader patterns of social stratification that shape not just

access to the virtual world, but also full participation in other common forms of information and communication technologies. The results suggest that there is no need to try to explain the online gender gap, for example, by theories specific to this type of technology, such as women's supposed "computer phobia," attitudes toward computers in the classroom, or the lack of nonaggressive computer games and websites suitable for young girls. All this may, or may not, be true. But it turns out that these are flimsy general explanations for why fewer women than men are online, because women are also less likely than men to have access to technologies delivering mass entertainment such as cable TV and VCRs. Even in affluent societies, poorer families lacking such common consumer durables as cable TV, VCRs, and automobiles are likely to lack Internet access as well.

In the longer run, technological and economic developments may well alter the market for Internet access, reducing costs, simplifying technical skills, and thereby widening usage. Generational trends are likely to be particularly important in terms of the long-term patterns of use, as younger groups eventually replace the older population. Public policy initiatives such as wiring all schools are likely to contribute to ameliorating some of the major disparities in computing skills and knowledge. Nevertheless none of these developments is likely to alter long-established patterns of social stratification in the short-term. Moreover, even if the Internet eventually reaches 85 to 95 percent of the population, there are still multiple layers of access and use. TV sets essentially come in only a couple of flavors. The box can be bigger or smaller, analog or digital, single speaker or home theater, cheaper or more expensive, and so on, but basically a TV is a TV is a TV, making no demands beyond how to switch the power button, with perhaps 100+ channels but still nothing to watch. In contrast, levels of Internet access can vary substantially. Today people living in poorer neighborhoods may be able to surf the Web from public libraries, schools, and community centers, or even cyber cafés, but this is not the same as having automatic access via high-speed connections at home and the office. Nor is it the same as having all Internet, all the time, downloadable via personal appliances, digital assistants, and cellular phones for today's wired road warriors in planes, ships, and trains. Internet technology is not going to stand still as long as the market continues to demand ever smaller, faster, and improved forms of delivery. Subsequent chapters go on to explore the underlying reasons for the social inequalities established here, and the broader implications for civic engagement and political participation in the digital age.

PART II

The Virtual Political System

Early in the next millennium your right and left cuff links or earnings may communicate with each other by low-orbiting satellites and have more computer power than your present PC. Mass Media will be redefined by systems for transmitting and receiving personalized information and entertainment. Schools will change to become more like museums and playgrounds for children to assemble ideas and socialize with children from all over the world. The digital planet will look and feel like the head of a pin.

Nicholas Negroponte, *Being Digital* (1995)

Politics on the Internet is politics as usual conducted mostly by familiar parties, candidates, interest groups and news media.

Margolis and Resnick, *Politics as Usual* (2000) p. vii.

Not To Be Taken Out
Reserve Reading Room